CELL TRACKING

BY

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THE CELL PROGRAM
EVERY CLAIM, WHAT IS BEHIND IT, AND WHAT IT WANTS

BIOHUB_CELL_TRACKING
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1

THE WORKBOOK IS THE LEDGER

Purpose: One place where every idea the cell program rests on is written down beside what is behind it. A reader then knows at a glance which claims are proved, which are measured, which are built but unmeasured, and which are still only theory.

Scope: `workbooks/cell_tracking/`. The thought experiment that sets the sieve on the cell program is in `thought_experiments/cell_tracking/`. The machine the program runs on is the engine workbook, `workbooks/engine/`.

1.1 HOW AN ENTRY IS RETAINED

Every claim carries one status, and the status says what backs it:

status	what it means
proved	an exact check ran over the whole set and found no exception; the count is given
measured	a number was taken on named samples; the number is given, and whether it survived
built	the code exists and runs; no measurement yet says whether it helps
theory	stated, not built
refuted	measured, and the measurement went against it; retained, with the number, and not tried again blind
not so	fails on its own arithmetic or in any machine, before anything is measured; retained, with the reason

A claim changes status only when a run changes it, and the run is named. Nothing is deleted from the ledger. An idea that failed stays, with the number that failed it.

1.2 ENTRIES

file	what it holds
cell_tracking_table.md	the cell program's n-body problem part by part: the physics, what the program does for each part, what it wants, every hypothesis tried with its result, the driver's stages, and the tracker's equation as reads of the machine
on_the_engine.md	what the engine's workbooks measured on the cell program's samples, and the sieve's sections that set it on the cells
build_plan.md	the order the cell program's wants are built in, and every ruling on them
ledger.md	every measurement of the tracker and its driver, in the order it was taken, with its samples and its result
scan_then_sort.md	the rule that orders the driver: every frame of every sample scanned first, sorting only at the end
walk_back.md	the rule every tracker stage must satisfy: it can be walked back to what it came from
build_right_first.md	the rule against building an interim version already known to need redoing

2

THE CELL TRACKING TABLE

Purpose: The micro n-body problem cell_tracking exists to solve, part by part. For each part: the physics we hold it to, what the program does about it, what it wants to do, every hypothesis tried there with its result, and the next move.

Scope: the program: `cell_tracking/src/`, the rules in `cell_tracking/base.cfg`, the cell-specific modules still under `engine/` until they move (`bodies`, `score_sample`, `answer_key`, `measure/*`, `group_objects`, `link_objects`, `relate_frames`), the physics in the thought experiments (the harmonics source above all, `proposal.md`, in `thought_experiments/cell_tracking/`; the rest are in `thought_experiments/engine/`), and the numbers in ledger.md. The machine the program runs on is a separate concern with its own table, engine_table.md; a row here names the machine rows (M1 to M12) it reads. Statuses follow README.md. A hypothesis with no run in the ledger says so.

2.1 TWO TABLES, TWO CONCERNS

The engine is the machine: exact arithmetic, the residual operator, the component tree, the moments, the one correlation operator, the marginal, the record machine, the files. The program is cell tracking: what a cell is, which body it is next frame, when it divides or dies. Each table is optimized on its own terms. They are optimized against each other only through the interface at the end of each table (what this program asks of the engine, what the engine offers). Merged, neither can be optimized on its own. A second program, RSNA knee MRI, runs on the same engine. The machine must stay free of cells for it to run.

2.2 THE PROBLEM

A developing zebrafish tissue is an n-body system: cells are the bodies, suspended in a fluid medium, touching and adhering through their membranes, moving, dividing, dying, entering and leaving the volume. We see it only as 100 frames of a 64×256×256 volume, with each sample's own deterministic noise on top. From that we must recover every body, its state, its motion, which body it is in the next frame, and its lineage. Everything must be exact, and no number may be chosen.

The score pays for three things: nodes matched to ground-truth nodes within 7 μm, edges between matched nodes, and divisions (0.1 of the score). Every row below is judged by what it moves on that score.

2.3 THE DRIVER: SCAN EVERYTHING, SORT AT THE END

"you scan everything and then you sort stuff. You don't need to sort it before you scan in all of the samples." The current driver partitions every frame as it reads it: the cut at ℓ^* (row 1) fixes each voxel's body from one frame before the sample's other 99 frames are seen. That is the moment deciding membership, which row 1's and row 9's physics forbid. The score reads only node positions (within 7 μm) and edges, never regions. So the new driver first scans every frame of every sample into points and fields that bound nothing. Only then does it sort: link, retain, divide, and grade.

A peak here is a point: a positive voxel of the residual that no one of its 26 neighbors exceeds. Nothing is assigned to it. The regions around peaks (the watershed's catchment regions, found by tobogganing, each voxel following its steepest neighbor) bound the set, and the driver does not form them. A body's shape is read from the field at its peak instead of from a region: the residual's exact second differences there are the $\ell = 2$ of its local field, with its long axis and aspect. Everything in this table is tracking and lives in `cell_tracking/`. The engine is called only for math: the residual, the drift, the entropy history, and one-to-one matching. Every run and every test is a job on the tessera schedule; nothing touches the device outside it.

"The whole point of the tracker is to track cells. If you can't walk back something you did you built it wrong." Every stage retains what it came from, and every decision retains what it replaced:

- a point retains its frame, voxel and exact level;
- a transform retains its exact inverse;
- a link retains its two points and its evidence;
- a choice in the sort retains the options it passed over, with their weights.

Nothing is deleted. What the sort does not retain is marked, and every mark can be walked back to the readings. A map that sends two readings to one (a clip, a rounding, a bin) cannot be walked back and is not built.

The organism physics each stage answers to is the thought experiments' (the harmonics source, `proposal.md`, and the two fluidic ones), row by row in the table below:

- the gradients define themselves, and nothing is tuned;
- a body's identity is the limit of its coherence over the whole sample instead of a tag;
- every body's $\ell \leq 2$ harmonics are its own, from intrinsic differences (mass density, membrane elasticity, aspect ratio), and oblate bodies make $\ell = 1$ and $\ell = 2$ dominate;
- the field moves as a whole (drift) and each body moves on top of it; a body's velocity is its dipole's time derivative, its next place is predicted from that before anything is matched, and v_origin = $-\nabla(\partial C_{1m}/\partial t)$ points back to where the motion started;
- bodies touch and adhere through their membranes, move together, and two membranes cannot pass through one another;

- a split is a boundary discontinuity; a division's daughters part along the parent's long axis with its mass conserved, and the local entropy bumps and settles;
- a lysis is a bump that rises into the floor and does not settle;
- mitosis and lysis change the fluidics for a great distance around them: a disturbance runs out through the tissue, overlapping disturbances are told apart by their harmonics, and each is traced back to its origin;
- oblate bodies deforming, snapping or throwing a pseudopod shed predictable dipole waves along their axis;
- what decays to max entropy is the floor; what holds low and moves is a body;
- every event is judged over the whole sample, never in the moment.

So. The set's intensities

WHAT IT DOES Every sample is read once, and the set's readings are counted into one exact histogram. Each sample gets its own affine footing, (reading − lo, hi − lo), with lo and hi converged from their 1% and 99%. The contrast is the set's cumulative count. Both are walked back exactly, and nothing is clipped (O2). Each sample's counts and footing go into its `.readings`, and the set's total, width and samples into `scan.set`. The count itself, C, goes into every `.points`, and each level walks back from its own file

ORGANISM PHYSICS (ROW) contrast across the set; the medium stays E1's to subtract (3)

OVER the whole set, before the scan

READS `engine_iapx_load`

LIVES IN `cell_tracking/src/scan/` (`scan_set`, its first pass)

STATUS built. **Measured** on the 25: 10,485,760,000 readings, and that makes the contrast 34 bits. Every footing converged in 6 to 16 steps, lo 34 to 699 and hi 330 to 2,099. Every `.points` holds the C rebuilt from the 25 `.readings`: `points_check` (`cell_tracking/src/check/points_check.c`, its own reader, apart from the scan's) rebuilds the readings, bits and C from the summed `.readings` and finds them in `scan.set` and in every head, 25 of 25, 0 faults

S1. Peaks

WHAT IT DOES every positive local the majority of the residual, as a point (t, z, y, x) with its exact level and the faces that cut its 26-neighborhood (O1, O4)

ORGANISM PHYSICS (ROW) a body is above the medium (1); the medium is subtracted whole (3)

OVER each frame, as it is read

READS `engine_residual_planes` (E1): So's contrast as planes of its width, with the residual's width returned

LIVES IN `cell_tracking/src/peaks/`, run by `cell_tracking/src/scan/`

STATUS built; peaks at 1, 9, 10 and 12 limbs **proved** against an exact host scan (131 checks, 0 failed). **Measured** on the 25: 6,518,615 points, 1,145 to 6,031 a frame, and 942,418 of them (14.5%) have a face in their neighborhood. Each level is 10 limbs. The residual takes 20.1 ms a frame and the peaks 0.86 ms. `points_check` reads every `.points` and finds 0 levels not positive, 0 voxels out of order and 0 outside the frame in the 6,518,615 points. `cell_tracking/test/check_test.sh` shows each of its 16 faults fires. The faces are counted, not yet stored per point (O4)

S1b. Shape

WHAT IT DOES at each peak, the residual's six exact second differences: the local field's $\ell = 2$, its long axis and aspect

ORGANISM PHYSICS (ROW) the body's own harmonics (2); oblate bodies (2, 9)

OVER each frame, as it is read

READS the frame's residual

LIVES IN `cell_tracking/src/peaks/`

STATUS built (`cell_tracking/src/peaks/shape.cu`): the scan writes each point's six second differences and its faces to the sample's `.shape` (`scan.cu`). `cell_tracking/test/shape_test.sh` holds its checks; its result is not recorded

S2. Drift

WHAT IT DOES the lag that best carries frame t's positive set onto t+1's

ORGANISM PHYSICS (ROW) the field moves as a whole first (4)

OVER each frame pair, as it is read

READS the engine's drift (E3, A4)

LIVES IN the engine; called from the scan

STATUS built, and the scan calls it: each frame's positive set (`cell_tracking/src/scan/drift.cu`) goes to the engine's `shift_agreement`, and the lag and its agreement are written to the sample's `.drift` (`scan.cu`), which the sort reads (S5)

S3. History

WHAT IT DOES per voxel and bit, the flips over the whole sample

ORGANISM PHYSICS (ROW) the floor and what holds low (12); the bumps of division and lysis (9, 10)

OVER the whole sample

READS `engine_entropy_set` (M8)

LIVES IN the engine; `--run entropy`

STATUS built

S4. The set and the null

WHAT IT DOES each sample's peaks in its null frames (row 8's draws), and what the set's samples say together, read before any sample is sorted

ORGANISM PHYSICS (ROW) a body is real where it is above the field-noise reading (8)

OVER every sample

READS S1 to S3

LIVES IN `cell_tracking/src/`

STATUS theory

S5. Motion, then links

WHAT IT DOES each point's next place is predicted first: the drift, plus its own velocity (its last link's displacement, the dipole's time derivative), plus the local flow (its neighbors' departures from the drift, weighted by level). Its candidates are the gate (O6): its nearest point of t+1, and every point of t+1 whose nearest prediction it is, with ties retained. The cost is the weighted squared distance from the prediction, $\|\cdot\|^2_w$ with $w = \sigma \circ \sigma$, weighed by the sample's own count of first-rank links at that distance (O7), plus the distance between the two shapes (S1b, O8), the shape taken from the chain's last point to retain a fate change from breaking the chain (H2). The first pass is one to one by `heaviest_matching` on exact integers. Two neighbors' links may not cross: the sign of $(c_A - c_B)\cdot(c_A' - c_B')$ is retained

ORGANISM PHYSICS (ROW) motion (5); local flow (4); identity by harmonics (2, 7); membranes do not cross (6)

OVER the whole sample, after its scan

READS S1, S1b, S2; the engine's matching (M7)

LIVES IN `cell_tracking/src/`

STATUS built in part (`cell_tracking/src/sort/`, `--run sort`): the prediction is the point's place, the drift and its own motion (its chosen link's, less the drift then); the gate is O6's; the cost is the weighted squared distance alone (the shape term is clear, and O7's weighting and the local flow are not built); the pass is one to one by `heaviest_matching`; crossings are counted instead of refused. Its links on 44b6_0113de3b, 44b6_0b24845f and 44b6_0c582fdc: 646,179, scored 0.063648 with every point retained (S11). `cell_tracking/test/sort_test.sh` holds its checks; its result is not recorded

S6. Chains and identity

WHAT IT DOES Linked points form chains, solved once over the whole sample as a min-cost flow (O10). Every cost is an exact integer on one footing: links from O7; "no link" from the null draw; appearance free at a face or the first frame. The local moves (O12), repeated to a fixed point, check the solve. A chain is retained where it is above the same chains in the null frames, over the whole sample, with no length chosen (O17). A chain that ends and one that starts at its predicted place, any number of frames on, are one chain: the gap is bridged (O14). Chains that part and rejoin, or a point ending beside one starting, are one body with two tops (O5, O13, O15). Each link's marginal is taken over its support component (O18), and each source retains its heaviest outcome (O19)

ORGANISM PHYSICS (ROW) identity is coherence (7); the null (8)

OVER the whole sample

READS S5, S4

LIVES IN `cell_tracking/src/`

STATUS theory

S7. Division

WHAT IT DOES A chain goes on as two: two points in t+1 whose nearest prediction is the parent's (O6), parting along the parent's long axis (S1b), with the history bumping at t and settling (S3). This is the evidence O9 learns. The daughters need not be alike: mass is conserved on their sum, and each takes its own harmonics (H1). The split sits at the frame of the strongest evidence along the chain, and a daughter chain that does not are above the null is no daughter (O16). The neighbors' departures from the drift in the frames after point back to it

ORGANISM PHYSICS (ROW) a split is a discontinuity; mass and axis; the bump that settles; the disturbance (9, 4)

OVER the whole sample

READS S5, S6, S1b, S3

LIVES IN `cell_tracking/src/`

STATUS theory

S8. Lysis

WHAT IT DOES a chain ends where the history bumps and does not settle, and its neighbors' departures point back to it; a chain that ends with no bump is a missed detection, not a death

ORGANISM PHYSICS (ROW) the bump that stays; the disturbance (10, 4)

OVER the whole sample

READS S6, S3

LIVES IN `cell_tracking/src/`

STATUS theory

S9. Faces

WHAT IT DOES a chain that starts or ends where its prediction leaves the view entered or left, exactly, with no buffer (O4, O11)

ORGANISM PHYSICS (ROW) entering and leaving (11)

OVER the whole sample

READS S5, S6

LIVES IN `cell_tracking/src/`

STATUS theory

S10. Disturbances

WHAT IT DOES every division and lysis (S7, S8) is an origin; the departures around it over the following frames are resolved by their $\ell = 1$ and $\ell = 2$ about each origin, which tells overlapping disturbances apart and traces each back; they then correct S5's local flow, and the links are sorted again

ORGANISM PHYSICS (ROW) mitosis and lysis change the fluidics far around them; dipole waves; retrograde tracing (4)

OVER the whole sample

READS S5, S7, S8

LIVES IN `cell_tracking/src/`

STATUS theory

Order of building: So (O2's norm, once the set is ingested); S1 (with O4's faces), S1b and S2 in the scan; then S5 with O6's gate, and S11 with every point retained (the first scored run); then O7's weights, S4, and S6 as O10's one solve with O12 to O15; then S7, S8 and S9 (O9, O16, O11); then S10 with O21's flow. O18 and O20 come in beside S6 and S11.

2.3.1 *OrganoidTracker's techniques, into the driver*

These were read from `vendor_for_ref/OrganoidTracker/` (read only), following `organoid_tracker_predict_positions.py`, `_predict_divisions.py`, `_create_tracks.py`, `_marginalization.py` and `_callibrate_marginalization.py` into the modules they call. Their pipeline runs in this order:

1. A network calls positions.
2. Nearest-neighbor candidates are drawn.
3. Networks score each link, and each position's chance of dividing.
4. One min-cost flow runs over the whole movie (dpct).
5. Local repairs fix what the flow missed.
6. Each link gets a marginal, and errors are flagged.

Each technique is listed with its chosen numbers, then our form of it: exact, nothing chosen, and in the sort after the scan.

O1. Peak calling

ORGANOIDTRACKER (WHERE) `position_predictor.py` (`peak_local_max`); `loss_functions.peak_finding`

WHAT IT DOES THERE A voxel is a peak where it equals the max over a (3, 13, 13) box and exceeds 0.1 of the range. At inference: min_distance 6 px and threshold 0.1, after interpolating 5 layers between z planes (`reconstruct_volume`). The peaks are called on a near-isotropic grid. Every voxel that ties its box max is a peak.

OURS: EXACT, NOTHING CHOSEN S1 as built: the 26 neighbors, no box, and no threshold past > 0. The z anisotropy enters the sort's norm, $w = \sigma \circ \sigma$ instead of interpolated planes. A tie goes to the lower index.

STAGE S1

STATUS built

O2. Intensity normalization

ORGANOIDTRACKER (WHERE) `position_predictor._split_into_patches` (the division and link predictors do the same)

WHAT IT DOES THERE Per frame, the readings are mapped onto [0, 1] between the 1% and 99% quantiles taken over the network's time window, then clipped. An optional top-hat (a Gaussian blur, then a minimum and a max filter of `tophat_mask`) is subtracted.

OURS: EXACT, NOTHING CHOSEN Taken, for contrast, once the set is ingested. Every sample is read once, and the set's readings are counted into one exact histogram, one count per 16-bit value. The bounds lo and hi are its quantiles: each is the least value whose cumulative count reaches its share of the set, exactly. Their clip is not built, because it cannot be walked back: every reading past a bound becomes the bound. What is built comes in two parts, both walked back exactly. (a) **The footing:** each sample's own affine map, (reading − lo) over (hi − lo), is stored as that pair and never divided. E1 is linear and cancels a constant, and this map moves no peak. It puts samples of different brightness on one footing for the set's readings (S4). The pairs are per sample. One pair for the whole set would change nothing, since every sample already shares the raw scale. They are read once the set is ingested. (b) **The contrast:** the set's cumulative count, $v \mapsto \#\{\text{readings of the set} \leq v\}$, is exact histogram equalization. It rises strictly on every value the set holds, and its inverse is the table of those values. It needs no levels, and it changes the scan's peaks the way the clip meant to. It carries the contrast ("cumulative count sound like more information"). For (a), the levels are not written in ("we can let the numbers converge on them"). They start at theirs, 1% and 99%, and iterate to a fixed point on the set. The rule (built; 6 to 16 steps on the 25) is that each bound is the two-means split of its half of the set: the readings below the median for lo, and those above it for hi. Each is iterated from its seed until it stops moving. The class sums and counts are exact integers, a reading is compared against the midpoint by cross-multiplying, and a one-dimensional two-means stops in finitely many steps. The top-hat is not taken: the residual subtracts the medium (E1).

STAGE S0

STATUS built

O3. The time window

ORGANOIDTRACKER (WHERE) the position, division and link networks

WHAT IT DOES THERE Each network sees frames t + dt over its window, a stack of frames.

OURS: EXACT, NOTHING CHOSEN The scan reads one frame at a time and retains everything. Every judgement that needs other frames is made in the sort (S4 to S10), over the whole sample, with no window.

STAGE S4 to S10

STATUS the design

O4. The border

ORGANOIDTRACKER (WHERE) `custom_filters.get_edges`; `links_postprocessor.postprocess`

WHAT IT DOES THERE Training ignores voxels within (2.5, 11, 11) px of the x and y faces and the last z plane. Positions within `margin_xy` of the edge are dropped.

OURS: EXACT, NOTHING CHOSEN Each point records which faces cut its 26-neighborhood, exactly and with no margin. S9 reads the record.

STAGE S1, S9

STATUS the record built with S1 (942,418 of the 6,518,615 points have a face in their neighborhood); S9, which reads it, is theory

O5. A dividing cell called twice

ORGANOIDTRACKER (WHERE) `division_predictor.remove_division_oversegmentation`

WHAT IT DOES THERE Two points both called dividing and within 4.5 μm of each other (1.5 × that if both are dividing) become one point at their midpoint. In the frame before, points within 4.5 μm of a dividing point, other than its nearest, are removed.

OURS: EXACT, NOTHING CHOSEN Two peaks of one frame stay two points, and the sort decides. Chains that part and go on are a division (S7). Chains that rejoin are one body with two tops (E9's reconvergence, S6). The saddle between two neighboring peaks is an exact number: the level of their LCA in the component tree (low fruit 6).

STAGE S6, S7

STATUS theory

O6. The candidate gate

ORGANOIDTRACKER (WHERE) `nearest_neighbor_linker.nearest_neighbor`

WHAT IT DOES THERE Each point of t+1 goes to the points of t within tolerance × its nearest distance, at max 5 of them; each point of t goes to t+1 the same way; the candidates are the union. The link network uses tolerance 2. A point whose place in the other frame is outside the image is skipped.

OURS: EXACT, NOTHING CHOSEN From each point's prediction (S5): its nearest point of t+1 under $\|\cdot\|^2_w$. From each point of t+1: its nearest prediction. The gate is the union, with ties retained. Two points of t+1 whose nearest prediction is the same point are that point's division candidates (S7). A prediction outside Ω is a leaving (S9). No tolerance and no count.

STAGE S5

STATUS built in S5's gate (`cell_tracking/src/sort/sort_link.cu`): forward and backward, each the least weighted length with ties retained, and their union; a prediction outside the view is retained and flagged. The division candidates (S7) and the leaving (S9) are not built

O7. The link's weight, learned without ground truth

ORGANOIDTRACKER (WHERE) `nearest_neighbor_linker.nearest_neighbor_probabilities`

WHAT IT DOES THERE Every point's shortest out-link is taken as true and its others as false. The links are counted in log-distance bins of 0.25, and a logistic curve is fitted to the counts. The penalty is $-\log_{10}$ of the odds.

OURS: EXACT, NOTHING CHOSEN The same count, with no bins and no fit. $\|\cdot\|^2$_w from the prediction is an exact integer. At each value d^2 the sample's (or the set's) count of first-rank links over all links is an exact rational, P(first | d^2). The count needs the whole sample scanned before any link is weighed.

STAGE S5, S4

STATUS theory

O8. The link network

ORGANOIDTRACKER (WHERE) `link_predictor`

WHAT IT DOES THERE Its inputs: patches at both ends over the time window, and the displacement given as CoordConv channels. Its output is Platt-scaled against curated links.

OURS: EXACT, NOTHING CHOSEN What it learns from is the displacement and the two ends' look. S5 reads the displacement from the prediction exactly, and S1b's pair of shapes for the look. Nothing is calibrated.

STAGE S5, S1b

STATUS not built

O9. The division network

ORGANOIDTRACKER (WHERE) `division_predictor`

WHAT IT DOES THERE Gives each point a probability from a patch over its window, Platt-scaled; the penalty is $-\log_{10}$ of the odds. It is trained on the mother's last frame (or 3 frames before to 2 after the split), with dying cells displayed more often.

OURS: EXACT, NOTHING CHOSEN The evidence is read from the data: the shape's aspect along its long axis (S1b), the history's bump (S3), and two t+1 points sharing one nearest prediction (O6). The frames around a split are where the bump rises and settles, read from S3 instead of a window.

STAGE S7

STATUS theory

O10. The global solve

ORGANOIDTRACKER (WHERE) `dpct_linker` (Haubold, Ales, Wolf, Hamprecht, ECCV 2016 [1]: generalized successive shortest paths; or Magnusson)

WHAT IT DOES THERE One min-cost flow over every frame at once. A node costs 2 if left unused and 0 if used. Appearance and disappearance cost their penalties, and are free at the first and last frames. A node may send two units (a division) where three things hold: its division penalty is under 2, it has more than one future, and its second-best out-link beats its disappearance. Five weights (link, detection, division, appearance, disappearance), all 1. A link is offered only within 4 of the best in and out penalties, and under 4.

OURS: EXACT, NOTHING CHOSEN S6 is this solve: once over the whole sample, after the scan. Every cost is an exact integer on one footing: a link from O7; a point's "no link" from its null draw (S4: the nearest it finds in a frame where no correspondence exists), as the marginal already weighs "none" (row 7). Appearance is free where the back-prediction leaves Ω (S9) and at the first frame. A division costs by O9's evidence. No weights and no cut-offs: O6's gate is the pruning. The machine's `heaviest_matching` (M7) solves one frame pair, one to one. The flow over the sample, with two-unit sources, is the tracker's own, in `cell_tracking/src/`. Whether a generic min-cost flow belongs in the engine is Doug's call.

STAGE S6

STATUS theory

O11. Appearance at the faces

ORGANOIDTRACKER (WHERE) `dpct_linker.calculate_appearance_penalty`

WHAT IT DOES THERE Within 8 μm of any face but z = 0, the chance of appearing rises linearly from 0.01 to 0.51. A random ±0.05 is added to every penalty "to help with optimalization".

OURS: EXACT, NOTHING CHOSEN S9: a start is an entering exactly when its back-prediction lies outside Ω, with no buffer. Ties break exactly (least index, least $\|\cdot\|^2$_w), never by noise.

STAGE S9

STATUS not built

O12. Local moves

ORGANOIDTRACKER (WHERE) `links_postprocessor.finetune_solution`, 4 passes

WHAT IT DOES THERE Drop a link where appearance + disappearance is cheaper. Reconnect a loose start or end by breaking another track, when the total drops. Add a daughter where the division plus the link (plus the other's disappearance) is cheaper. Swap two crossing links. Drop a division whose penalty is over 2 by cutting the worse daughter.

OURS: EXACT, NOTHING CHOSEN The same moves, as exact integer comparisons of the same costs, repeated until no move lowers the total: a fixed point with no pass count. On an exact whole-sample solve they change nothing. They are the check that it is optimal, and the repair where the gate cut an option. The swap is the no-crossing rule (S5, row 6) seen from the cost side.

STAGE S6

STATUS theory

O13. A loop from over-detection

ORGANOIDTRACKER (WHERE) `links_postprocessor.connect_loose_ends`

WHAT IT DOES THERE A track ends and another starts within 4 frames. If the link + 2 < disappearance + appearance, they are joined and the points between are deleted (—====— → ———).

OURS: EXACT, NOTHING CHOSEN A chain that parts and rejoins is one body with two tops over that stretch, whatever its length (E9's reconvergence). The null decides whether it is retained, and no penalty enters.

STAGE S6

STATUS theory

O14. A missed frame

ORGANOIDTRACKER (WHERE) `links_postprocessor.bridge_gaps`

WHAT IT DOES THERE A loose end at t and a loose start at t+2, each among the other's 6 nearest within 10 μm and neither with a nearer loose partner. If 2 < disappearance + appearance, a point goes in at t+1, at their midpoint. A point with a strong division score and one future counts as a loose end too.

OURS: EXACT, NOTHING CHOSEN S6's gap bridge: the end's prediction is carried across any number of frames; the pair is mutual nearest; and the inserted point is the exact midpoint, stored as a sum and a count and written as a half voxel where odd. No distance or count is chosen.

STAGE S6

STATUS theory

O15. A duplicate in one frame

ORGANOIDTRACKER (WHERE) `links_postprocessor.bridge_gaps2`

WHAT IT DOES THERE A loose end and a loose start in the same frame, within 7 μm and each the other's nearest. The end is a duplicate: its past is linked to the start, and it is removed.

OURS: EXACT, NOTHING CHOSEN Two points of one frame, one ending and one starting, mutual nearest: one body with two tops.

STAGE S6

STATUS theory

O16. The division's frame

ORGANOIDTRACKER (WHERE) `links_postprocessor.pinpoint_divisions`, `remove_spurs_division`

WHAT IT DOES THERE A division moves one frame later when a daughter's division penalty is at least 1 below the mother's. A daughter that ends at once is deleted when the mother's division penalty is positive.

OURS: EXACT, NOTHING CHOSEN S7 puts the split at the frame of the strongest evidence along the chain. A daughter chain that does not are above the null is no daughter.

STAGE S7

STATUS theory

O17. Clean-up

ORGANOIDTRACKER (WHERE) `remove_tracks_too_deep` (both ends past z = 25), `remove_tracks_too_short` (6 points or fewer, not reaching the last frame), `remove_single_positions`

WHAT IT DOES THERE Removes the tracks and points so described.

OURS: EXACT, NOTHING CHOSEN Not taken as written: every length, depth and margin there is chosen. A chain is retained where it is above its null (S4, S6).

STAGE S6

STATUS not taken

O18. Marginals

ORGANOIDTRACKER (WHERE) `local_marginalization_functions`; `organoid_tracker_callibrate_marginalization.py`

WHAT IT DOES THERE For each link, every flow-consistent state of the subgraph within 3 hops is weighted $10^{\wedge}-E$, and P = Z_link / Z. The hop count drops while the estimated count of states exceeds 2^{16}. Options that leave the subgraph are folded into the node's end (`combine_events`, $1 - \Pi(1 - p)$). A division enters as a node's second unit. A temperature of 1.5, fitted by Platt scaling against curated links, scales the energies. Links with |penalty| $\geq$ 4 use only the target's in-links.

OURS: EXACT, NOTHING CHOSEN Built as the machine's marginal (M6) with the program's weights (row 7's `--marginal`): exact integers, the masses cancel, no temperature. Over the new driver's points, the local set is the support component (low fruit 2). Nothing is outside it, and nothing is folded. Divisions are two-target arrangements (low fruit 3).

STAGE S6

STATUS the marginal built (M6); over points, not built

O19. Dropping uncertain links

ORGANOIDTRACKER (WHERE) `organoid_tracker_marginalization.py`

WHAT IT DOES THERE A link with a marginal under 0.99 is dropped, and with it every out-link of a mother. Where the division score is high and no division was made, a second daughter is added at (x + 1, y + 1).

OURS: EXACT, NOTHING CHOSEN Each source retains its heaviest outcome in its marginal (an argmax, no threshold). No point is invented.

STAGE S6, S11

STATUS theory

O20. Error flags

ORGANOIDTRACKER (WHERE) `cell_error_finder`, `errors.py`

WHAT IT DOES THERE Each point is checked for: a track end before the last frame; no past after the first; more than two daughters; a merge (two pasts); a low division or link score; a short cell cycle; a missed division (a high score, one future, and not high in the next two frames); moving too fast.

OURS: EXACT, NOTHING CHOSEN The graph's own invariants, checked before the output: in-degree $\leq$ 1; out-degree $\leq$ 2, and 2 only with a division verdict; every end a death, a leaving or the last frame; every start an entering, a daughter or the first frame. They are truthy records, counted per sample. A speed or cycle limit written in would be a chosen number. Both are read from the set's distributions.

STAGE S11

STATUS built in part in S11: in-degree $\leq$ 1 and out-degree $\leq$ 1 checked (nothing divides yet), the ends and starts counted per frame, H4 confirmed on 297 of 297 frame pairs; explaining an end or a start waits on S7 to S9

O21. Local flow

ORGANOIDTRACKER (WHERE) `particle_flow_calculator`

WHAT IT DOES THERE The mean displacement of the neighbors within ±50 px in x and y and ±2 planes in z, each to its single next or previous point, skipping divisions and ends.

OURS: EXACT, NOTHING CHOSEN S5's local flow: the neighbors' departures from the drift, each weighted by its level. The neighbors are the point's neighbors in the Delaunay graph of the frame's points under ‖·‖_w, with no radius, and divisions and ends are skipped as there. The first-pass links give it, and S10 sorts again.

STAGE S5, S10

STATUS theory

O22. Image offsets

ORGANOIDTRACKER (WHERE) `core/images.ImageOffsets`, set by hand in the offset editor; `is_inside_image`

WHAT IT DOES THERE A per-frame offset of the image. Positions are retained in its frame, and in-view tests read it.

OURS: EXACT, NOTHING CHOSEN S2's drift is the offset, read from the data per frame pair (E3).

STAGE S2, S9

STATUS built in the engine

2.3.2 *Hawkins et al. 2025, into the driver*

Source. Hawkins, Jones, Wesselman, Cesa, Moeller, Wingert, "Mathematical modeling reveals cell differentiation processes and progenitor kinetics necessary for proper nephrogenesis", *Front. Cell Dev. Biol.* 13:1695380 (2025). Doug's upload, read.

The model. ODE models of the zebrafish pronephros from 10 to 96 hpf. Two progenitor pools (anterior G, posterior J) feed three tubule populations (anterior A, posterior P, multiciliated M). Every rate (proliferation, fate, death) is a Gaussian in time, K(t) = K_max · 2^(−((t − K_time)/K_width)²). The models are fitted by bounded BFGS, simulated annealing and Monte Carlo over 100 × 100 grids of starting counts, and compared by AIC and likelihood-ratio tests.

The measurements.

- PH3 (G2/M): about 7.5% of PST cells and 3% of DE cells at 20 hpf.
- Caspase-3 (apoptosis): near 0 in both.
- Zebrafish cycles from the literature: 6 to 10 h, with G2/M about 1 h, in retina and hindbrain at 28 hpf; 2 to 24 h for progenitors, by tissue and stage.

The finding. The best model needs three fates from a progenitor:

- symmetric division (two progenitors);
- asymmetric division (one progenitor and one mature cell);

- a fate change without division (the cell matures without dividing).

What it gives the tracker is physics, with none of its numbers written in:

#	the physics	what it changes in the driver	stage
H1	Asymmetric division: the two daughters are different cells.	A division does not need its daughters alike. Mass is conserved on their sum (row 9's D_mass already reads $\beta(m_c + m_s) = \beta(m_p)$), and they part along the parent's axis. Their shapes (S1b) and levels may differ, and from then on each daughter's chain carries its own harmonics.	S7
H2	A fate change without division: a cell changes what it is without dividing.	A chain goes on through a change of shape or level with no split, because identity is coherence (row 7), not a constant look. S5 weighs a shape against the chain's last point, never its first. A gradual change does not break the chain. A change of harmonics with no second point in the gate is a fate change instead of a new body.	S5, S6
H3	Death is near zero in the early pronephros (Caspase-3 about 0 in PST and DE at 20 hpf), and a few percent of cells are in G2/M at a time.	This agrees with S8: a chain that ends away from a face, with no bump, is a missed detection instead of a death. Divisions are rare against continuations, as the ground truth shows (87 of 199 samples hold any division, at max 5). The rates are read from each sample's own counts after the sort, never written in.	S8, S7
H4	The population balances: its change is births minus deaths plus arrivals.	On any output with in-degree ≤ 1 and out-degree ≤ 2, $N(t+1) - N(t) = D(t) - E(t) + S(t+1)$ exactly. D counts the two-daughter points of t, E the points of t with no future, and S the points of t+1 with no past. The physics says every end is a death, a leaving or the last frame, and every start is an entering or the first frame. The ends and starts left unexplained are the verdict, counted per frame and per sample.	S11
H5	Each process's rate varies in time and peaks once (their Gaussian K(t)).	The sample's division count per frame is read as a curve over the whole sample after the sort: a set-wide reading beside S_4's, with no shape fitted to it.	S4, S7
H6	Hypotheses are compared by likelihood ratio (AIC, LRT).	Ours is the null (S_4). Each event (a division, two tops, a missed frame) is retained where its exact count is above the same count in the null frames. No χ^2 and no parameter penalty.	S4, S6

2.4 THE TABLE

1. The bodies

THE PHYSICS WE HOLD IT TO A body is above the medium, whole; the field carves its boundary, and no gradient is tuned.

DOES The program takes the machine's component tree (M3) and one cut: the level with the max components, proved against the forest's count. `grow_leaves` makes the leaves. `group_objects` joins leaves into objects by touch in one frame, transitively; on the current bodies nothing touches, and every object is one body (row 6). Since the split, the slide and

the overlap are machine operators over probes and probe links (M3). The program turns ground-truth cells and divisions into probes and links and grades the census itself, in `cell_tracking/src/slide/slide_score.cu`. That file is rewritten to the probe interface and is **not yet compiled or hooked in**: it hooks into score_sample's frame loop, and score_sample has to move to cell_tracking first.

WANTS One node per cell. 44b6 is estimated at about 260 cells a frame and 6bba at about 700; the tree gives far more pieces.

TRIED, AND WHAT IT GAVE Every node: **measured**, 5.8 M nodes against 880,906 estimated, score 0.279. The 400 largest a frame: **measured**, 0.661 on the 25 44b6. `is`: **measured**, 0.12 on the older dump (38,114 nodes retained). `merge_split`: on by default, no A/B. `dish`, `cohere`, `accrue`, `agree`: no run in the ledger. The slide (`--slide`, before the split), **measured**. Every level where the ground-truth cells' sharing changes is found by bisection over the levels. Each probe's component count is proved against the tree's count at that level: 0 of 26,454 differ. On 6bba_48816121, the cut holds 931 of 935 ground-truth cells, and 50 are alone on their leaf. At each frame's best single level, 903 are alone; at some level of their own chain, 925. The max-components level sits low in the tree: many small pieces, with the cells merged into a few giant blobs. Frame 0: the cut is at level 84,314 of 1,135,213; all 4 ground-truth cells are alone from 1,045,988 (204 components) down to 463,871. The overlap at the same residual (`--overlap`), **measured**. Frames t and t+1 are cut at the same exact residual r. At the drift, each component's distinct partners are counted from the exact (earlier root, later root) pairs; mutual is the AND, both ends with exactly one. The sampled r values are each frame's ground-truth partition events and its cut. On 6bba_48816121 (99 pairs, 3,615 cuts), the earlier frame's cut has 61,679 components, of which 15,979 overlap exactly one and 6,688 are mutual (11%). The best mutual over each pair's cuts sums to 13,783. Mutual peaks high in the tree, at 150 to 200 components a frame with 70–80% of them mutual. Frame 0→1 at level 1,045,988: 204 components and 144 mutual; at the cut: 329 and 29. On 44b6_0113de3b, the cut has 39,507 components and 7,962 mutual (20%). The event census, **measured**. It classes every component: 1→1 move, 1→2, 2→1, 1→0 and 0→1. Mutual counted from either side agrees on every cut. On 6bba_48816121 at the cut, about 70% of the components have no partner at all. Of the ground truth's 5 divisions, 0 are parted at max-components cut. At each pair's best sampled cut, 5 are present, 3 have their daughters apart, 2 are parted (the parent's component overlaps both daughters' components) and 1 is caught as 1→2. Daughters separate only where the parent is gone, and one global drift misses daughters that move apart. Ground-truth cells per leaf (`--box-history`), **measured**. On 6bba_48816121, 881 of the 931 ground-truth nodes that land on a leaf share it with another ground-truth node, on 110 leaves. Up to 13 ground-truth cells sit on one leaf of 779,789 voxels, and every ground-truth division's parent sits on a leaf of 652,575 to 1,004,159 voxels whose box is the whole 64×256×256 view. On 44b6_0113de3b, 0 of 52 share. So ℓ^* cuts 6bba into a few giant components that hold many cells each. The 6bba edge scores (856/918 on this sample) count leaf-to-leaf links, and the majority of those leaves hold many cells. They do not measure per-cell tracking.

STATUS built; the node count is the score's lever; on 6bba the cut holds many cells a leaf

NEXT Bodies, not pieces, as the nodes: row 2's fingerprint and row 9's entropy decide membership. Hook `slide_score.cu` in once score_sample moves. The machine's LCA identity (M3, theory) gives every level where two ground-truth cells part without bisection: one query per neighboring pair of ground-truth cells.

2. Each body's intrinsic state: its spherical harmonics

THE PHYSICS WE HOLD IT TO Every body has unique harmonics because of intrinsic physical differences (mass density, membrane elasticity, aspect ratio). Cells are oblate, and for an oblate body $\ell = 1$ (dipole) and $\ell = 2$ (quadrupole) dominate. Up to $\ell = 2$ the harmonics carry exactly what the moments carry: $\ell = 0$ the mass, $\ell = 1$ the index sums, $\ell = 2$ the six second moments.

DOES The machine accumulates every body's exact moments (M4). `grow` carries them to `Tree Frame.sums` and `.moments`. `coherence` exports them to the viewer. The print (`fingerprint_program`, 52 steps) writes each body's $\ell \leq 2$ fingerprint as a record: the mass band and six signed golden bands of K'/m^2, $K' = s_a s_b(m{\cdot}M - S_a S_b)$ scaled by `voxel_pm` over its common unit. `print_pair` weighs two prints against each other. **The linker (E8 in the equation below) does not read the print yet.** The sums are also read by `parallax`, `arc` (`track_bend`), `velocity`, `division` and the export.

WANTS The print in the linker's pool, to tell a cell from its neighbor. $\ell > 2$ needs the body's surface: theory. Oblate spheroidal harmonics in place of spherical: theory.

TRIED, AND WHAT IT GAVE The print: **measured**, 81,807 bodies of 44b6_0113de3b in 11.7 ms, CRC-64 `590f3a2c386647e3` twice, 0 differ from the host. Linking by it: see row 7 (`--print-match`).

STATUS built, read only by `--print-match`

NEXT Put the print into E8's pool: waits on Doug (build plan, Waiting on Doug 3).

3. The medium: the fluid the bodies sit in

THE PHYSICS WE HOLD IT TO The wide term is the medium's scattering, its turbidity, the fluid's mean entropic flux. It is noise to us and is subtracted whole, in exact two's complement, and the medium resolves to zero (phase cancellation).

DOES The program passes the machine's residual (M2) one smooth order and one background order per axis, {2, 34, 34} and {6, 96, 96} (`SMOOTH_ORDERS`, `BACKGROUND_ORDERS`, track.h). **They are a debt**: they came from `n = 4(σ/v)²` rounded to even, with two chosen scales, σ = 6/5 µm (smooth) and 2 µm (background) (`for_orior/stripped/src/exact_track.py`). The rounding and the chosen σ both break the rules.

WANTS Orders read from the data, per sample, with nothing rounded. Doug's rulings, the per-axis coherence reading gives an exact integer period P in voxel steps, and **order = period** (n = P). Flatten takes a smooth and background pair per sample (not built). The medium's

physics now has an exact form (M2's heat identity): the residual is 4^m times what m explicit heat steps of the medium remove from the smoothed field. "The medium's mean entropic flux" is literally the subtracted term.

TRIED, AND WHAT IT GAVE The key against the passes, and the unit sweeps: see M2 (proved exact; the sweeps 2× faster), with its numbers on the samples in on_the_engine.md.

STATUS orders chosen and rounded (debt)

NEXT Build the per-axis coherence reading (machine, M2's next), then orders per sample. Whether the tracker runs the unit sweeps by default (`--unit-sweep`) is Doug's call.

4. Microfluidics: the medium's flow

THE PHYSICS WE HOLD IT TO The field moves as a whole (drift) and each body moves on top of it; the whole moves first. Mitosis and lysis change the fluidics for a great distance around them: a division or a lysis sends a hydrodynamic disturbance through the surrounding tissue. Oblate bodies shed predictable dipole waves when they deform, snap or throw out a pseudopod.

DOES Bulk drift only: the machine's drift (M5, `shift_agreement`) gives the one lag that best carries this frame's set bits onto the next frame's, `lag_to_next` (the demon's eyes). Nothing reads a local flow, a disturbance or a wave.

WANTS The disturbance field around each event read from the residual, which lets an event far off be felt where it arrives. Overlapping disturbances untangled by their harmonic coefficients, each traced back to its origin, retrograde.

TRIED, AND WHAT IT GAVE `motion_check` (drift run twice, every count compared): built, no disagreement recorded. `retain_view` (no drift removed): no run in the ledger.

STATUS drift built; flow theory

NEXT Build plan C3: each body's departure from the drift (v*_b − d_t), summed over its contacts and weighted by mass, then traced back toward the division and lysis events it came from.

5. Each body's motion

THE PHYSICS WE HOLD IT TO A body's velocity is the time derivative of its dipole. Its next place is predicted from that before any overlap is counted, and the dipole axis points back to where the motion started: v_origin = $-\nabla(\partial C_{1m}/\partial t)$.

DOES The climb (M5, `climb_machine`): each body climbs from the drift to its own lag, one step in 26 directions at a time, by how many of its voxels agree, until no step helps. `--velocity` writes the dipole's change per body as a record beside the climb (see the verdicts). **`Engine Body.velocity` exists and nothing writes it; the linker does not read the velocity record.**

WANTS Velocity from the dipole's change across frames, used to predict each landing, and checked against the climb.

TRIED, AND WHAT IT GAVE `sticky`, `mass` (land where the majority of the body lands), `spiral` (golden spiral offsets beyond the climbed lag), `arms` (the same climb at gaps 2 to n, by which time bodies too close one frame on have parted): all built, no run in the ledger. `--velocity` (the `velocity` record program): the dipole's change is written per body as an exact rational, (S′m − Sm′)/(m·m′) per axis, beside a truthy verdict that the climbed lag is on a lattice point next to it. **Measured** on 44b6_0113de3b, device equal to the host on every lane: of 35,755 bodies with a climbed forward, the climb and the dipole agree on all three axes for 8,720 (z 22,660, y 13,227, x 13,306); the last change predicts the next climb for 752 of 35,966. On 6bba_48816121, 6bba_09961292 and 6bba_cdcfe533, **measured**: the check agrees on all axes for 2,144 of 21,366, 22,027 of 60,714 and 5,466 of 29,252; the prediction for 201 of 21,724, 2,666 of 58,840 and 497 of 28,743; 0 differ from the host.

STATUS climb built; dipole velocity built and measured on four samples

NEXT The leaves are pieces (818 a frame against about 260 cells), and a piece's dipole moves with the cut instead of the cell: measure the same on bodies, and on the 25.

6. Protein interactions: contact and adhesion

THE PHYSICS WE HOLD IT TO Cells touch and adhere through membrane proteins; bodies in contact move together and pull on each other. The membrane is a scale of the problem (`membrane_pm`, 4 nm, the bilayer's thickness).

DOES **No two bodies touch on the current engine.** The bodies are the separate components of the cut at ℓ*, and two components cannot share a face without being one. So `grow_leaves` holds no touching pairs (`joined_count` = 0), and `group_objects`, `sticky`, `web` and `--contact-side` have nothing to read. The joined leaf pairs came from the older basin engine. The volume faces each body touches (`touches`) are recorded. `voxel_pm` is read by the print, `division` and `contact_side`. `membrane_pm` and `species` are read into the `.cfg` and written back out, **with no reader**.

WANTS Contact as a force-free constraint: bodies adhered in one frame stay adhered unless the data parts them. The membrane scale in place of any voxel count. Membership never by touch alone (row 9).

TRIED, AND WHAT IT GAVE `sticky`, `web`: built, no run in the ledger. Grouping by touch: **measured** on the older basin engine to chain up to 2,256 pieces into one object at the widest; on the current engine every object is one body. `--contact-side` on 44b6_0113de3b and three 6bba: **measured**, 0 touching pairs on every sample, because there are none.

STATUS contacts not stored on the current bodies; adhesion theory

NEXT What "touching" is for components of a cut is Doug's to define: for example, the component tree's next level down, where two bodies first join, or a gap of the membrane's width.

The tree gives the first of these exactly: two bodies first join at the level of their LCA (M3). Until then the no-crossing verdict (built, `--contact-side`) has no pairs. B3's deformation waits on C1: the membrane deforms, and a body and its match overlap only in part. What stays inside the match under every jitter is the body's internal constituents, and their shift is the velocity and vector.

7. Correspondence: which body it is next frame

THE PHYSICS WE HOLD IT TO Identity is the limit of coherence: a body is who it is because its coherence carries on, and a tag plays no part. One-to-one where bodies are one-to-one.

DOES The overlap triples (M5, `body_overlap`) count shared positive voxels per pair at the drift. `pick` retains the branch sharing the max, `unbound` takes the nearest center by the swept vector (the jitter step it meets at over its magnitude), and `resolve` rejoins reconverging branches by union-find. The machine's one-to-one matching (M7, `heaviest_matching`) is wired in by `--print-match`: the `print_pair` record program weighs every candidate (the overlap triples and the climbed forward) by how far its seven-band print lies from the body's, and the heaviest one-to-one set rewrites the forwards. `--marginal` writes each body's exact context-aware probability $P(b \to b' \mid G) = Z_{b \to b'}/Z$ over its local set, with the machine's marginal (M6) and the program's weights (see the verdicts). It reads the triples and the null, and the linker does not read it yet. **Measured** (internal count): 98.4% of 6,358 edges on the 25 44b6, 97.0% of 3,873 on five 6bba.

WANTS Link each body to the one whose fingerprint agrees (row 2), one-to-one by `heaviest_matching` on fingerprint cost.

TRIED, AND WHAT IT GAVE Under the metric: **measured**, 87.1% hit on 44b6 (10.9% land on a neighbor, 5.75 μm) and 78.5% on 6bba (19.6%, 6.7 μm). `--print-match` on 44b6_0113de3b: **measured**, 56,927 pairs swept at once in 217 to 245 us, 31,130 of 80,697 leaves matched one to one, 19,113 forwards changed, edges unchanged at 38/12 of 50, because base.cfg's `unbound` linker picks by overlap weight and center cost and reads the forward only as one more candidate. `--marginal` on 44b6_0113de3b (with the floor's null): **measured**, 34,297 bodies asked in 50 to 58 ms (30,444 in their context, 3,853 alone, 8 too wide), 123,538 options, device equal to the host. Of 49 ground-truth edges asked, the max probable in context is the truth for 16 and "no link" for 27. The heaviest candidate alone is the truth for 38, and the climb's forward for 35. Where a target wins, it is the truth 16 times of 22. The null's weight dominates: it is the null climb's held count, which the climb maximizes, while T is counted at the drift. Putting the two on one footing is Doug's to set. With Doug's transform (`--box --marginal`, the links weighed by T*, C at each body's own climbed lag): **measured**, 51,214 bodies asked in 240 ms, 195,658 options, device equal to the host. Of 49 ground-truth edges, the max probable in context is the truth for 19 and "no link" for 24; the heaviest candidate alone gets 37 and the climb's forward 35. "No link" is weighed by the best of the floor's null draws ($\max_k h^0_k$). Whether "the null draw's fraction" means the best draw, each draw as its own outcome, or something else is Doug's. No run in the ledger for `share`, `cast`, `parallax`, `arc`, `settle`, `damp`, `vote`, `mutual`, `forest`, `merge_target`, `forward_only`, `focus` or `tower`.

STATUS built; losses are neighbors

NEXT Row 2's test. The marginal's local set G is one hop (the sources competing for b's candidates, and their candidates). M6's factorization says the exact marginal over the whole frame equals the marginal over b's connected component of the support graph, and over nothing smaller: take G as that component, and the probability is the global one exactly (theory).

9. Division: mitosis

THE PHYSICS WE HOLD IT TO A split is a boundary discontinuity, judged over the whole sample and not the moment. One body's fingerprint becomes two whose masses add to it and whose dipoles part along the parent's quadrupole axis. The local entropy bumps and settles into two lows.

DOES `assign_bodies` (only when node rows are asked for) marks a body SPLIT when nothing lands on it and its back landing is a body that went elsewhere, and gives it that body's id as parent. `--division` writes the mass and axis verdicts for every parent and two pieces whose backward climbs land on it (the `division` record, 100 steps). `assign_bodies` does not read them yet, and there is no entropy check.

WANTS Two linked children confirmed by mass conservation, the parent's $\ell = 2$ axis, and an entropy bump that settles.

TRIED, AND WHAT IT GAVE Divisions matched: **measured**, 0 on the older dump. The division term earns 0. `--division` on 44b6_0113de3b: **measured**, 142,649 triples in 13.8 ms, mass conserved on 3,356, split along the long axis on 99,155, both on 2,452, 0 differ from the host. That sample's ground truth holds no division. Of the 199 samples' ground truth, 87 hold any division, at max 5 (6bba_48816121). On 6bba_48816121, 6bba_09961292 and 6bba_cdcfe533: **measured**, 326,836, 85,146 and 99,724 triples, each in 1 to 3 ms, 0 differ from the host. The three samples' ground truth holds 13 divisions. Only 3 put the parent and both children on three separate leaves, and 2 of those are among the triples. Mass holds on 1 of the 2, the axis on 2, both on 1. Why the rest are lost, **measured**: 10 of the 13 have both children on one leaf. One frame after the split, the cut at ℓ^* has not separated the daughters. On 6bba_48816121 (`--box-history`), the parent and daughters of all 5 of its divisions are on leaves of 0.65 to 1.0 M voxels whose box is the whole view (row 1). None has a node off every leaf, and none falls outside consecutive tracked frames. On the 3 with three separate leaves, 4 of the 6 children climb back to their parent. So the loss lies in the cut (row 1), and the climb holds.

STATUS fate built; mass and axis verdicts built and measured

NEXT The daughters inside one leaf are likely separate components higher in the same component tree. One cut level for the whole frame cannot hold both a just-divided pair and everything else. That is row 1's selection (Ultrack's [2] one node per nested chain; build plan, Waiting on Doug 5). The census's $1 \to 2$ class (M3) equals the set of sources a division arrangement would add to the marginal (row 7). Then build plan C2 (the entropy bump).

10. Death: lysis

THE PHYSICS WE HOLD IT TO An uncontained dissipation: the entropy bump rises into the floor and does not settle, and the body bleeds into the background.

DOES `assign_bodies` marks VANISHED (nothing onward, not at a face), ABSORBED (landed in a body another carried) and MERGED. Nothing tells a lysis from a missed detection.

WANTS A lysis told from a vanishing by its bump rising into the floor.

TRIED, AND WHAT IT GAVE The box history (C2's measurement): **proved** on 44b6_0113de3b (81,807 boxes × 9 windows × 16 bits, 0 of 11,780,208 differ from walking every box, 5.4 s) and 6bba_48816121 (0 of 9,021,600, 6.3 s). Every ground-truth division's parent on 6bba_48816121 sits on a leaf whose box is the whole view (row 1). Its history is the view's, and a bump cannot be read there.

STATUS measured, no verdict

NEXT Build plan C2. `--box-history` holds each body's box (`climb_machine_extents`) and gathers its flips per window and bit from one running-sum key a window and bit, by 8 corners. The verdict (a bump that rises and settles, or rises and stays) waits on row 1's cut and on Doug: which bits, and how the windows (11 transitions) meet a one-frame event.

11. Entering and leaving the volume

THE PHYSICS WE HOLD IT TO Bodies cross the volume's faces; a body that appears at a face entered, one that disappears at a face left.

DOES ENTERED and LEFT from `touches` (the volume faces a body meets).

WANTS Nothing further now.

TRIED, AND WHAT IT GAVE none

STATUS built

NEXT none

12. The noise floor: what is not a body

THE PHYSICS WE HOLD IT TO The noise is deterministic, constant and unique per sample: shot, thermal and read, fixed pattern, quantization. It is read, never modeled. Entropy separates it: what decays to max entropy is floor -4, and what holds low and moves is a body.

DOES The machine's entropy history (M8) with its window-overlap cloud; the `.cfg`'s `floor` positions it down first.

WANTS Every departure from the floor placed, and only those places asked about (the demon's waveform collapsing onto U | U). The four noise keys imprinted and stamped over the set in one cycle.

TRIED, AND WHAT IT GAVE Anchors: **measured**, bits 0 to 4 in about 46% of frames everywhere, anchors in bits 6 to 11, none common to every sample. History: **measured**, bits 0 to 3 at 499 to 500 per mille in 19 of 25. A constant region in 6 samples: measured, cause unknown. Direction: measured, each sample's own.

STATUS measured

NEXT Build plan C4 (waits on Doug's design), then membership by entropic direction (rows 1, 9, 10).

13. A body's edge

THE PHYSICS WE HOLD IT TO The edge stays with the body under every jitter of the view; what falls in and out is the floor at its boundary, set aside and not averaged.

DOES The jitter sweep (`LINK_SWEEP_STEPS`, a box a voxel wide doubling out past the view) exists and is used only to score a link's disagreement. The jitter core (M5, `climb_machine_core`, `--core`) is built and measured.

WANTS Every boundary voxel scrubbed under the sweep, leaving a body's edge and mass its own and not one frame's cut.

TRIED, AND WHAT IT GAVE The core per width on 44b6_0113de3b, **measured**. Proof: the width-0 core mass equals C at the climbed lag on 35,755 of 35,755 forward climbers. Of 161,708 climbers (both sides, adjacent frames), 72,186 reach width 0, 33,827 width 1, 22,884 width 2, 10,230 width 3, 606 width 4 and 2 width 5. Core mass is 90,205,526 / 36,098,925 / 6,542,528 / 307,434 / 4,956 / 2. Which axis ends each core is not yet measured. Mutual pairs whose core displacement lies within half a voxel of the climbed lag on every axis: 10,003 of 16,176 at w=0, 4,409/10,410 at w=1, 2,578/8,727 at w=2, 456/3,413 at w=3, 15/117 at w=4. The deeper the core, the less its motion is the lag's. 49.6 s, with D3 sharing the GPU.

STATUS core built and measured

NEXT Build plan C1 (every width, retained per width, between frames): a voxel of a body stays at width w when every offset in the box of half-width w around the climbed lag lands it inside the match. Mass, sums and moments re-summed per width, and the print summed over the edge that stays.

14. Size: growth and division by ratio

THE PHYSICS WE HOLD IT TO Sizes spread on a golden ladder, since growth and division scale by ratios.

DOES The machine's golden ladder (M11, `band_of`), read by `damp`, by `group_objects`, by the residual survey, by the print and by `division` (mass conserved is $\beta(m_c + m_s) = \beta(m_p)$). `--mass-band` (`band_or_count`) compares the bands instead of the counts wherever two sizes are compared: `group_objects`, `arm_landing`, `focus_links`, `assign_bodies` and `track_bend`. Exact sums stay counts, since a band there would round.

WANTS A size read on the ladder instead of any count.

TRIED, AND WHAT IT GAVE `damp`: no run in the ledger. `--mass-band`, D3 on the first 25: identical to counts on every one of the 24 that ran both sides, with 6,134 edges and 5,176 correct / 227 wrong / 2 none / 729 missed. It is identical by construction: on the scored path every reader is gated off. `track_bend`, `arm_landing` and the scorer's member choice compare within one object's members, and every object is one leaf. `group_objects` needs joined pairs, and there are 0. `focus_links` needs `--focus`; `assign_bodies` needs nodes output. The 25th, 44b6_551a5dba, loaded in the counts run and then fails its .iapx CRC in the bands run and on every later read, with the file's modification time unchanged (for Doug).

STATUS built, inert

NEXT Live only once objects hold more than one leaf, or once bodies touch (row 6).

15. The whole state, once

THE PHYSICS WE HOLD IT TO The field's state is a set of bodies at every time, and one exact file holds all of it.

DOES The machine holds the field (M9: `.kcr`, `.knf`, `flatten`, `.bapx`). No file holds the bodies' fingerprints.

WANTS A file type for the bodies and their $\ell \leq 2$ fingerprints, written once, for linking to read bodies and not voxels.

TRIED, AND WHAT IT GAVE See M9, and `flatten`'s numbers on 44b6_0113de3b in on_the_engine.md.

STATUS field stored by the machine; bodies' prints not stored

NEXT With row 2.

2.5 THE TRACKER'S EQUATION

The tracker, written out as the one function it computes, stage by stage, from the code as it runs under `cell_tracking/base.cfg` (`pick`, `unbound`, `merge_split`, `resolve`, `climb: machine`). Every quantity is an exact integer unless a line says otherwise. Each stage names the code, and the machine operator it reads (M-rows and A-sections of engine_table.md). The algebra can be worked on the formula and checked against the code.

2.5.1 *Notation*

- t is a frame, x = (z, y, x) a voxel of the view $\Omega = [0, D) \times [0, H) \times [0, W)$, and I_t(x) the frame's 16-bit reading.

- w = (w_z, w_y, w_x) = (16, 1, 1) are the axis weights (`AXIS_WEIGHTS`, track.h). $\|v\|^2_w = 16v_z^2 + v_y^2 + v_x^2$. They equal $\sigma \circ \sigma$, with σ = voxel_pm / gcd(voxel_pm) = (1,625,000, 406,250, 406,250) / 406,250 = (4, 1, 1), the same σ `contact_side` and the print already scale by (see the low fruit below).
- · is 1 when its statement holds and 0 otherwise.
- β(n) is n's rung on the golden ladder (A-section A9).

2.5.2 *E1. The residual (row 3): the machine's A1*

R_t = 2^g · (B_s ∗ I_t) − B_b ∗ (B_s ∗ I_t), with the program's orders s and b (row 3's debt). The positive set is P_t(x) = [R_t(x) > 0].

2.5.3 *E2. The bodies (row 1): the machine's A2 and A3, then `grow_leaves`*

- The cut is ℓ^*_t = min { ℓ : C_t(ℓ) = max C_t }, the lowest level with the max components (A2). The choice of one level for the whole frame is the program's.
- The bodies B_t are the components at ℓ^*_t, with mass m_b, index sums S_b, second moments M_b, peak p_b and touches (A3). The centroid is carried as the pair (S_b, m_b) and never divided.

2.5.4 *E3. The drift (row 4): A4, `shift_agreement`*

d_t = argmax_v Σ_x P_t(x) · P_{t+1}(x + v)

2.5.5 *E4. The climb (row 5): A4, `climb_machine`*

For each body b of frame t, A_b(v) = Σ_{x∈b} P_{t+1}(x + v). The climb starts at v_0 = d_t, steps to the neighbor u (26 of them) with the largest A_b(u) while A_b(u) > A_b(v_k), ties to the least $\|u\|^2_w$, and stops at v^*_b.

- The forward is f(b) = L_{t+1}(p_b + v^*_b), the body of t+1 under the peak, or none.
- The climb's held score is h(b) = A_b(v^*_b).
- The backward runs the same from t+1 to t, starting at −d_t, and gives back(b′) and its lag $v^*_\{b'\}$.

2.5.6 *E5. The null (row 8): the same climb toward a frame where no correspondence exists*

$h^0_k(b) = A_b^{(t \to t_k)}(v^*)$, one draw k per floor on the identity line when a floor is positioned down

2.5.7 *E6. The overlap triples (row 7): A4, `body_overlap`*

T(b, b′) = |{ x : L_t(x) = b, L_{t+1}(x + d_t) = b′ }|, retained where T > 0

2.5.8 *E7. The objects (row 6): `group_objects` (`merge_split`)*

b ~ b′ ⇔ b, b′ touch ∧ (O_{t − 1}(back(b)) = O_{t − 1}(back(b′)) ∨ f(b) = f(b′) ≠ none)

The objects O_t are the transitive closure of ~, one frame after another. **On the current bodies, E7 changes nothing**: two components of one cut cannot share a face. `grow_leaves` writes no touching pairs and every object is one body ("largest 1" on every run).

2.5.9 *E8. The link (row 7): `link_objects_unbound`*

For an object O of t and each object Q of t+1 that O's triples or forwards reach:

W(O, Q) = Σ_{b∈O} Σ_{b′∈Q} T(b, b′)

A forward that reaches Q enters the pool with weight 0.

κ(b, b′) = min over the four pairings (lag or lead) of (s ≪ 40) | (Σ_a w_a (ℓ_a + ℓ′_a)² mod 2^{40})

- ℓ is b's forward lag and ℓ′ is b''s backward lag. Their sum ℓ + ℓ′ is how far the round trip misses.
- s = min { j ≤ 12 : 2^j ≥ max_a |ℓ_a + ℓ′_a| } is the jitter step it meets at.
- K(O, Q) is the least κ over the member pairs that reach Q.

  Links(O) = { Q : W(O, Q) = max_Q′ W(O, Q′) ∧ K(O, Q) = min over those Q of K }

The link is the heaviest overlap. The round-trip cost only breaks ties.

2.5.10 *E9. Reconvergence (row 7): `resolve`*

When O has two links whose single-link chains meet again at one node, every pair of nodes along the two chains is joined (union-find). The nodes are the classes that result.

2.5.11 *E10. Fate (rows 9 to 11): `assign_bodies`*

- SPLIT: nothing lands on the body, and its back landing is a body that went elsewhere. That body becomes its parent.
- VANISHED, ABSORBED, MERGED, ENTERED and LEFT come from the landings and `touches`.

2.5.12 *The verdicts built beside the path (records on the machine's record engine, M10; truthy: zero false, nonzero true; AND is a product, OR a sum)*

These are written for every body, pair or triple in one sweep each, device equal to the host. **E8 reads none of them yet.**

- **The print** (row 2, `fingerprint`), per body: β(m) and the six signed golden bands of K′ / m², where K′ = s_a s_b (m M − S_a S_b) is scaled by `voxel_pm` over its common unit.
- **The print cost** (row 7, `print_pair`), per pair: the distance between two prints. The ceiling 1,275 is a body against itself.
- **Velocity** (row 5, `velocity`), per body and its forward, and per axis:
  - Δ_a = (S′_a m − S_a m′) / (m m′), stored as the pair (numerator, m m′)
  - R_a = L_a · m m′ − (S′_a m − S_a m′)
  - c_a = COMPARE(m m′, |R_a|), and the verdict V_a = c_a (c_a + 1). It is nonzero exactly when |Δ_a − L_a| < 1, the climbed lag located on a lattice point beside the dipole's change.
  - V = V_z V_y V_x.
- **Division** (row 9, `division`), per parent p and two pieces c, s whose backward climbs land on p:
  - D_mass = 1 − |COMPARE(β(m_c + m_s), β(m_p))|
  - D_axis = [3 Δ^{T} K_p Δ ≥ tr(K_p) |Δ|²], where Δ = S_c m_s − S_s m_c
  - D = D_mass · D_axis
- **Contact side** (row 6, `contact_side`, two passes), per touching pair A, B of frame t whose forwards A′, B′ both exist:
  - pass 1: N = σ ∘ (S_A m_B − S_B m_A) per pair, where σ_a = voxel_pm_a / gcd(voxel_pm);
  - pass 2 reads two pass-1 records: dot = N · N′, c = COMPARE(dot + 1, 1) = sign(dot);
  - retained = c (c + 1) and crossed = c (c − 1).

  dot is (c_A − c_B)·(c_A′ − c_B′) in physical space times the positive m_A m_B m_A′ m_B′ / gcd², and its sign is exact. Two membranes cannot pass through one another, and a real pair retains its side. dot = 0 includes A′ = B′, the pair landing on one body.
- **The marginal** (rows 7 and 8), the machine's A5 with the program's weights: ω(b″, b′) = T(b″, b′) (or T*, below) and ω(b″, none) = max_k h^{0}_k(b″). P(b → b′ | G) = Z_{b→b′} / Z. Every source's weights share its own mass as denominator. The masses cancel, and Z and every Z_{b→·} are exact integers compared by cross-multiplying, without temperature, energy or calibration. OrganoidTracker 2.0's P(A | G) is the same sum over exp(−E/T) of calibrated network energies.

2.5.13 *The whole, in one line*

Tracks = Resolve(Links(W(T(B(R(I)), d)), K(v*(B, d), v*(B′, −d))), Objects(B, f, back))

This reads from the inside out:

1. the residual R;
2. its bodies B at the cut ℓ^*;
3. the drift d between frames;
4. the overlap T at the drift and the climbed lags v*;
5. the objects from touch and shared landings;
6. the link, by the heaviest W and then the least K;
7. the reconverging chains joined.

The records (print, velocity, division, marginal) are functions of the same B, v* and T. How they enter W and K is Doug's to set (Waiting on Doug, item 3, in the build plan).

2.5.14 *The tracker's stages as reads of the machine's one operator (23 September: "we are repeating the same operation close to 10 times")*

Almost every stage above is the machine's correlation C (A4) followed by a different reduction:

stage	the read of C
E2 moments m, S, M	`C_{t,t}[a, a](0)` weighted by 1, x, x x^T
E3 drift d	argmax_v Σ_{a,b} `C_{t,t+1}[a, b](v)`
E4 climb A_b(v)	Σ_{b′ ∪ ∅} `C_{t,t+1}[b, b′](v)`, climbed to a local max v*_b from d
E4 landing by mass	argmax_{b′} `C_{t,t+1}[b, b′](v*_b)`
the backward climb	the same box transposed: `C_{t+1,t}[b′, b](−v)` = `C_{t,t+1}[b, b′](v)`
E5 null h^0	Σ_{b′ ∪ ∅} `C_{t,t_k}[b, b′](v*)`, in a null frame
E6 overlap T	`C_{t,t+1}[b, b′](d)`
the transform T*	`C_{t,t+1}[b, b′](v*_b)`, with Σ_{b′} T* + T*(b, ∅) = h(b)
E8 link weight W	Σ over each object's members of T
C1 jitter core, every width	S_w[b, b′] = Σ_{‖u‖∞ ≤ w} `C_{t,t+1}[b, b′](v*_b + u)`: nested boxes around v*_b
contact κ_w (row 6)	Σ_v (B_w ⋆ B_w)(v) · `C_{t,t}[A, B](v)`: the box's autocorrelation, a tent, over the frame's own box
the slide and the census (row 1)	C at pairs of levels, a rectangle sum of the own-node table (A2)
velocity, division, print	the moments above, read in pairs and triples

What it replaces today: the drift computed once (`shift_agreement`, over every v, by an exact NTT); the climb again per body (27 shifts a step); the overlap again at the drift; the landing again (`land_mass`); the null again per draw; the core and contact each another pass; the census again per

sampled cut. The machine's box (`--box`) proved the climb, the null and the overlap are reads of the one C (engine table, M5; its numbers on 44b6_0113de3b are in on_the_engine.md). The window of shifts is the open design choice, and it is the program's to name in its request: the machine holds no scale.

2.5.15 *Doug's transforms, 23 September*

- **The null's exact transform.** The link's weight is the climb's own functional split by the target it lands on:

  $T^*(b, b') = |\{ x \in b : P_{t+1}(x + v^*_b) \wedge L_{t+1}(x + v^*_b) = b' \}|$, and summed over b', $\Sigma_{b'} T^*(b, b') + T^*(b, \varnothing) = A_b(v^*_b) = h(b)$

  where $\varnothing$ is a positive voxel in no body (R > 0 below the cut). In the marginal, $\omega(b, b') = T^*(b, b')$ and $\omega(b, \text{none}) = \max_k h^0_k(b)$. Both are counts of the same body's voxels under the same operator, and the masses still cancel.

- **Contact** (row 6). Each body has an assigned boundary. At width w, $\kappa_w(A, B) = |\{ x : d_A(x) \leq w \wedge d_B(x) \leq w \}|$, where $d_A(x) = \min_{y \in A} \|x - y\|_\infty$. It is nonzero exactly where the two boundaries overlap at w: the truthy contact, its size the magnitude.

- **The jitter core** (C1, rows 13 and 6). The match $b' = f(b)$ has $D_{b'}(y) = \min_{z \notin b'} \|y - z\|_\infty$. A voxel x of b stays at width w when $D_{b'}(x + v^*_b) > w$. Built and proved: the width-0 core is Cb, b', exactly, on every forward climber of 44b6_0113de3b. The widths end at 5 there; the data sets where they end, and the view bounds them at (shortest extent + 1)/2.

- **The cut** (row 1). ℓ^* slides through every level of the component tree, to the top, and each nested chain retains the node that stays coherent between frames. One level for the whole frame is replaced by one node per chain.

2.5.16 *Low fruit in the tracker's algebra (23 September)*

Each item is algebra on the equation above, exact, **theory until built**. Each names what it replaces.

1. **One scale vector.** $w = \sigma \circ \sigma$ with σ = voxel_pm / gcd(voxel_pm): the link's norm (E4, E8), the print's scale and the contact side's σ are the same vector. `AXIS_WEIGHTS` becomes derived, not written in, and any geometry (another microscope, the MRI's 0.3 mm against 3.5 mm) gets its weights exactly: the machine's `decimal_double` (A0) turns a spacing's decimal string into an exact integer, and the gcd makes them coprime integers. Replaces the constant `{16, 1, 1}`.

2. **The marginal over the component.** The exact frame-wide marginal factors over the connected components of the support graph (A5). b's probability is exact when G is b's whole component, and the current one-hop G is exact only when the component is one hop wide. Replaces the one-hop set; the "too wide" bodies (8 on 44b6_0113de3b) are the components past `MARGINAL_ARRANGEMENTS_MAX`, which the machine's permanent on the component (A5) measures instead of guessing.

3. **Divisions as arrangements.** A division is a source that takes two targets. The census's 1→2 class (A2) is exactly where that arrangement is needed. The marginal gains a two-target outcome only on those sources, and the rest retains the current injective sum.

4. **The slide by LCA.** Two ground-truth cells share a component at level r exactly when $r \leq$ level(LCA). Ordered by the tree's DFS order, k ground-truth cells need $k - 1$ LCA levels to give their partition at every level at once (A2). Replaces the bisection over levels.

5. **The overlap at every level from one count.** The census at every sampled cut is one own-node pair table summed over DFS ranges (A2). Replaces a recount per cut.

6. **Contact is an LCA level.** Row 6's "where two bodies first join" is level(LCA(A, B)), exact, with no gap to choose.

2.6 WHERE THE TRACKER IS NOT YET EXACT, OR HOLDS A CHOSEN NUMBER

where	what	in code
the residual's orders	{2, 34, 34} and {6, 96, 96} come from two chosen scales (6/5 μm, 2 μm) and $n = 4(\sigma/v)^2$ **rounded to even**. no rounding, anywhere; order = period from the per-axis coherence reading, per sample	`SMOOTH_ORDERS`, `BACKGROUND_ORDERS`, track.h; `exact_track.py`
w = (16, 1, 1)	equals $\sigma \circ \sigma$ with σ = voxel_pm / gcd(voxel_pm) = (4, 1, 1) exactly, but it is written in as a constant, not derived from `voxel_pm`	`AXIS_WEIGHTS`, track.h
κ's lead	(mine ± behind/2) / behind rounds by truncating division when frames are more than one apart; it is exact on consecutive frames (behind = 1), which is every frame	`leaf_disagreement`, relate_frames.cu
κ's magnitude	retained mod 2^{40}; it wraps silently above 2^{40}	`LINK_MAGNITUDE_BITS` 40
s	the jitter steps stop at 12, a chosen number	`LINK_SWEEP_STEPS` 12
the marginal's set	one frame step and one hop; the exact marginal needs the whole support component (low fruit 2). Every target is taken once. A division is not an arrangement yet (low fruit 3). The bound of 2^{16} arrangements is OrganoidTracker's, not measured here	`MARGINAL_ARRANGEMENTS_MAX`
the null in the marginal	ω(b, none) is the best null draw's retained count, a count of the body's voxels like T, but T is counted at the drift and h^0 at the climbed null lag	`score_best_null`, score_sample.cu
`cohere` (off in base.cfg)	a mean motion is divided by truncating division	group_objects.cu
the cut	one level (the max components) for the whole frame	`max_tree_objects` as the program calls it

2.7 WHAT THE PROGRAM ASKS OF THE ENGINE, AND WHAT IT OFFERS

The interface through which the two tables optimize against each other.

the program asks	the engine's row	state
probes and probe links, not ground-truth cells: the slide's partitions and the overlap's census	M3	built (engine side); `slide_score.cu` not hooked
the component tree's LCA levels and DFS order, to replace bisection (low fruit 4, 6)	M3	built: `max_tree_nodes`, `max_tree_probe_levels`, `max_tree_probe_partition`, proved on the host, 18071 checks, 0 failed; `slide_score.cu` not hooked, and the slide still bisects
one own-node pair table at the drift, summed over DFS ranges (low fruit 5)	M3, M5	built: `max_tree_pairs` and `max_tree_overlap_sums`, proved against direct counts in the same run; not hooked; M5 not built
a per-axis coherence reading that returns an exact integer period (order = period)	M2	theory
a smooth and background pair per sample in flatten	M9	ruled, not built
the permanent over a whole support component, and its size	M6	theory
exact spacings from decimal strings, for σ	M1	built (`decimal_double`)
a file for the bodies and their prints	M9	theory
the residual over readings of any width (So's contrast is the set's cumulative count, up to the set's reading count, far past 2^{16}). The engine's integers are exact, bit-packed and of any width. The request names the input's bits. The output's width is derived as input bits + Σ smooth orders + Σ background orders + 1 and returned, and peaks read that width	M2	built. `unit_sweep` takes limb planes of a declared width and refuses a value past it. `engine_residual_planes` returns the residual with its limbs, and `peaks_find` takes the limbs. **Proved**: the planes against the 16-bit path, 336 checks, 0 failed (engine table, M2), and the peaks at four widths, 131 checks, 0 failed. The scan runs it at 34 bits in and 10 limbs out

What the program offers the engine: nothing in the machine may know it. The measurements above are the engine's test load (44b6, 6bba), and the proofs the program runs (device equal to host, 0 lanes differ) are the machine's regression.

2.8 WHAT TO WORK TOWARD, IN ORDER

The order is the build plan's (build_plan.md):

1. **Built as records beside the path:** the print (row 2); the print cost with one-to-one matching (row 7, `--print-match`); velocity (row 5, `--velocity`); division mass and axis (row 9, `--`

`division`); the exact marginal (rows 7 and 8, `--marginal`); contact side (row 6, `--contact-side`); sizes on the ladder (row 14, `--mass-band`). E8, the link, reads none of them yet. How they enter W and K is Doug's to set.

2. **Move the cell-specific modules** to cell_tracking: `score_sample` and `coherence` are in `cell_tracking/src/`, and `vis_png` is in `view/`. Then hook `slide_score.cu` into the frame loop (row 1).

3. **Rows 6 and 13 (B3 with C1).** The jitter core per width.

4. **Rows 9 and 10 (C2).** The entropy bump that settles (division) or stays (lysis).

5. **Row 4 (C3).** Local flow from each body's departure from the drift.

6. **Row 3.** Orders per sample from the period reading, replacing the rounded debt; **row 12 (C4, waits on Doug).**

7. **D3.** Every want A/B on the 25, `above_null` (row 8) included, written into the ledger and this table.

3

THE CELL PROGRAM ON THE ENGINE

Purpose: Hold what the engine's workbooks measured on the cell program's samples, and the sieve's sections that set it on the cells. The engine workbook carries the machine alone and points here for the program's numbers.

Scope: text from the engine workbook: engine_table.md (M2, M5, M9, M10, M12 and M14), noise_sieve_tower.md (§14 and §15, and its lines on the program's node policies, links, random numbers, edge rows and floor) and tessera_scheduler.md (the driver's jobs). Each part names where it was, and its words are as they was there. Statuses follow README.md.

3.1 FROM THE ENGINE TABLE

3.1.1 *M2. The residual operator*

The unit sweeps against the key: **proved**, 0 of 419,430,400 lanes differ on 44b6_0113de3b's 100 frames and 0 on 400 more frames (6bba_48816121, 6bba_09961292, 6bba_cdcfe533, 44b6_0b24845f); 14.0 ms a frame against the key's 29.0 on 44b6_0113de3b, 14.3 against 28.1 on the four (**measured**). Confirmed again after every error-wiring build of proved on 100 frames, 0 lanes differ.

The cell program's scan runs it on the 25 at 34 bits in and 10 limbs out, 20.1 ms a frame (**measured**; cell table, S1).

3.1.2 *M5. The correlation operator C*

The box (`--box`) on 44b6_0113de3b: **proved**. C is read at the 27 shifts around every climbed lag and at the drift for 726,273 climbers (every forward pair and every null draw): 20,335,644 cells, 30,196,969 entries, 15.3 s. A cell meeting more labels than the kernel's table (10,507 cells) writes its raw pieces and the host merges them, with no cap. At the climbed lag, C sums to the climb's retained count on all 726,273; the climbed lag is the highest of its 27 neighbors on all 726,273; at the drift, C equals `body_overlap` on all 80,697 labels.

3.1.3 *M9. The files*

`flatten`: **measured**, 138 to 170 s on 44b6_0113de3b; the cause is its residual slab sized to half the free device memory: 107.0 s at that slab, 9.0 s one frame at a time, byte-identical output, every stage 10 to 20× slower under the big slab.

3.1.4 *M10. The record machine*

The programs themselves (print, velocity, division, contact side) belong to the cell program.

3.1.5 *M12. Errors and integrity*

Every wiring build checked on 44b6_0113de3b: 38/0/12 edges and 0 lanes differ, both modes.

3.1.6 *M14. The scheduler*

`track_driver` submits through it: `--ingest` is one job and each `--run` part is one; the signum is the part's name and the whole effective request; the declaration is the largest sample's lattice in 16-bit lanes, from the source's description for `--ingest` (`engine_source_lanes`, which reads no voxel) and from the `.iapx` head otherwise; the times are 2 s holding, 20 ms sweep and 5 s idle; a job that is not taken, or is withheld and lost, fails its part. **Measured** on 44b6_0113de3b, with the driver and the daemon built to `build/verify_driver` (exit 0) and run in a scratch `TESSERA_STATE`. The ingest is granted its declaration of 838,860,800 bytes and peaks at 5,091,037,184 (25,392 ms), sample root ad3d9846…, set root b68d522c…. Two proves in a row are each granted 838,860,800 and peak at 3,958,566,912 and 3,962,761,216, with the same roots. The history is then 128 bytes, two records and the seal, and the daemon ends once idle. A second run on the real state gives the same declarations and ingest peak, and the two prove peaks in the other order: identical requests measure 4,194,304 bytes (2^22) apart.

3.2 FROM THE TESSERA SCHEDULER: THE DRIVER SUBMITS (24 SEPTEMBER)

`track_driver` runs every job through tessera. `--ingest` is one job, and so is each `--run` part.

- **The signum** is the BLAKE3 hash of the part's name, a NUL, and the whole effective request. A changed setting gives a new signum.
- **The declaration** is the largest sample's lattice in 16-bit lanes. For `--ingest` it comes from the source's description (`engine_source_lanes`, which reads no voxel); for every other part, from the `.iapx` head.
- **The times** are 2 s holding, 20 ms sweep and 5 s idle.
- **The daemon** is `tessera_daemon` beside the driver, which the driver starts when none answers.
- **A part fails** if its job is not taken, or is withheld and lost; `--override` admits a withheld job on its declaration.

Measured on 44b6_0113de3b, in a scratch state directory:

run	declared and granted	peak measured
`--ingest`	838,860,800	5,091,037,184
`--run iapx-prove`	838,860,800	3,958,566,912
`--run iapx-prove` again	838,860,800	3,962,761,216

The history then contained two records and the seal (128 bytes), and the daemon ended once idle. A second run on the real state gave the same declarations and ingest peak, with the two prove peaks in the other order.

The two findings these runs gave are the scheduler's, in tessera_scheduler.md.

3.3 FROM THE NOISE SIEVE TOWER

3.3.1 *§6. The demon's arms: the node policies*

claim	status
binary probes in place of chosen thresholds	built; `is` gave 0.12 on the older components dump, far below taking every node (0.22), because it retained 38,114 nodes against 484,255 estimated

3.3.2 *§8. The three irreducible sets: true coherence*

set	in the engine	status
true coherence	the residual's structure above the medium: bodies and their links	built; 97.0% of 3,873 labeled edges linked on five 6bba samples by the internal count

3.3.3 *§10. A universal Turing machine and the demon*

claim	status
outcomes read deterministically, with no random number anywhere	built: no pseudo random number is formed anywhere in the tracker

3.3.4 *§13. The seed crystal: the edge rows*

claim	status
a flat field propagates nothing; a minimal fixed asymmetry gives every choice one direction	the Windows and Linux builds give byte identical edge rows over the 25

3.3.5 *§17. The floor positioned down first: the driver*

The `.cfg`'s `floor` section names where each sample's noise keys are stored (`--floor <dir>` on the command line).

claim	status
the floor section in the `.cfg`, positioned down before anything else runs	the driver positions the floor down first

3.3.6 *14. The sieve on this competition: splits, entropy, and each body's harmonics*

`proposal.md` sets the sieve on the Biohub volumes directly. It is the source nearest the score, and each of its mechanics has a concrete form here.

The engine holds nothing; gradients define themselves. No gradient is tuned and no threshold is chosen: the field carves the boundaries where the data puts them. This is already the engine's rule for its cuts. It is also the rule the node policies `is` and `above_null` were written to: a body is a node because it is here, not because it ranks in a chosen top n.

One precision note. The source speaks of a `10^ − 68` precision floor. The engine has no precision floor: its arithmetic is exact at every width it runs, and every width is proved before the run. The only floor is the measured one in the data (noise_sieve_tower §5).

A split is a boundary discontinuity, judged over the whole sample. A mitosis or a lysis is not decided in the frame it happens in. Over the sample's whole history there will be a bump in entropy at about that time, and the two differ: a mitosis is a clean fork where local entropy bumps and settles, while a lysis is an uncontained dissipation into the background. Both change the fluidics far around them. The metric pays for this: the division Jaccard is 0.1 of the score, and the engine earns none of it (0 divisions matched on the older components dump). The test is concrete. A body's fate is already written for every body (`bodies`: split, merged, vanished, absorbed). The entropy of the region around a candidate split, taken per frame from the exact residual, can say whether a real fork happened. Two linked children and a bump that settles are a division; a bump that bleeds into the floor is a lysis.

Each body's harmonics are its second moments. The source asks for each body's spherical harmonic coefficients, with the dipole ($\ell = 1$) and quadrupole ($\ell = 2$) dominant for the oblate shapes cells take. Up to $\ell = 2$, a body's harmonics carry exactly what its moments carry:

order	harmonics	what they hold	in the component tree, exactly
$\ell = 0$	one	the mass	`MAX_TREE_FIELD_MASS`
$\ell = 1$	three	the dipole: mass times the centroid	`MAX_TREE_FIELD_SUM_Z`, `_SUM_Y`, `_SUM_X`
$\ell = 2$	five	the quadrupole: the traceless second moment tensor, the body's oblateness and its axis	`MAX_TREE_FIELD_MOMENT_ZZ`, `_YY`, `_XX`, `_ZY`, `_ZX`, `_YX` (six terms; the trace is the sixth)

So every body the tree finds already carries its $\ell \leq 2$ fingerprint as exact integers, with nothing to fit. The source's "unique harmonics for each body because of intrinsic physical differences" is then a matching rule. A body in the next frame is the same body where its mass, dipole

displacement and quadrupole agree, up to the motion the frame shows. The retrograde vector $-\nabla(\partial C_{1m}/\partial t)$ is the time derivative of the dipole: the body's centroid velocity, run backwards to where the motion started.

Where it lands on the score. On the five 6bba samples, 19.6% of ground-truth edges are lost because a cell's link lands on a neighboring node, a median 6.7 μm from the right one (ledger). Neighboring nodes differ in mass and quadrupole even where their centroids are close, and a body's fingerprint tells it from its neighbors. A link chosen by the $\ell \leq 2$ fingerprint instead of by overlap alone is the direct test of this section.

claim	status
no tuned gradient or threshold	built: the cuts; the `is` and `above_null` policies
a precision floor of 10^ − 68	not so: the arithmetic is exact, with no floor
a split is a boundary discontinuity, confirmed over the whole sample by an entropy bump that settles; a lysis by one that bleeds out	theory; bodies' fates are built, the entropy test is not; the division term (0.1 of the score) is 0 today
mitosis and lysis change fluidics far around them	theory
each body's $\ell \leq 2$ harmonics stored exactly	built: mass, first and second moments per body in the component tree
linking bodies by their $\ell \leq 2$ fingerprint wins back the neighboring node losses	theory; the direct test against 19.6% on the five 6bba samples
spherical harmonics above $\ell = 2$	theory; they need the body's surface, not only its moments

3.3.7 *15. Edges by jitter, membership by sample coherence*

The engine's rule for what a body is, stated directly: **the jitter scrubs and oversamples to define a body's edges, and coherence over the whole sample assigns membership.**

Edges by jitter. A single frame's cut puts a boundary voxel on one side or the other by chance: the field noise at the edge decides it. The jitter sweep already in `relate_frames` (a box a voxel wide, doubling out past the whole view, `LINK_SWEEP_STEPS`) moves the view by every small offset and asks again. Scrubbing a body's boundary under every jitter oversamples it. The voxels that stay with the body under every offset are its edge; the ones that fall in and out are the floor at its boundary, read and set aside instead of averaged. This is noise_sieve_tower §5's floor applied at a body's surface.

Membership by sample coherence. Which pieces make one body is not decided in a frame either. A piece belongs to the body its coherence carries on with across the whole sample, the same way the harmonics source judges a split over the whole sample and not the moment (§14), and the same way identity is the limit of coherence (noise_sieve_tower §9). Pieces of one body cohere through every frame; pieces of two bodies part somewhere in it. So grouping by what touches in one frame, which runs transitively through touching cells, gives way to grouping by what coheres over all of them.

claim	status
the jitter sweep at every offset out to the whole view	built: `LINK_SWEEP_STEPS`, used today to score a link's disagreement

claim	status
a body's edge stays with it under every jitter	theory; the sweep exists, and its use on edges is not built

Never from one frame without context. Membership is never assigned from a single frame with no knowledge of the local entropy. A frame on its own cannot tell a body's edge from the floor beside it, or two bodies touching from one body. Once there is a history of entropic direction at a place (whether its local entropy is falling toward order, rising toward the floor, or holding), what each place is can be derived. A body is where entropy holds low and moves with it. The floor is where entropy sits at its max and goes nowhere. A split is a bump that settles into two lows, and a lysis is a bump that rises into the floor.

How it is measured, in the engine's terms. The anchor count already reads every frame of a sample once, per voxel and per bit. The same pass can carry the history at no extra cost. For each bit, it counts the frames whose bit changes from the frame before, the transitions, in windows of frames. A bit at the floor changes about half the time in every window. A bit inside a body barely changes while the body is there. A bit at a moving edge changes in a run, then settles. The windows in time order are the entropic direction at that voxel, as exact integer counts. No logarithm and no float is needed, because the direction is read from how the counts move instead of from their size.

claim	status
membership is never assigned from one frame without the local entropy	a rule of the engine; the current grouping breaks it (it groups by touch in one frame)
a history of entropic direction separates body, floor, edge, split and lysis	theory
the history is carried in the anchor pass, as windowed transition counts per voxel and bit, at no extra pass	theory; the anchor pass is built and its counts are on disk
membership assigned by coherence over the whole sample, not by touch in one frame	theory; the current grouping is by touch in one frame, transitive through touching cells

4

THE BUILD PLAN: THE CELL PROGRAM

Purpose: The order the cell program's wants in cell_tracking_table.md are built in, and every ruling on them. Each build takes the next item and not whatever looks nearest. Built in order, one item at a time; an item is done when its program runs on the device and its row below says so.

Two tables, two concerns: retain the cell table and the engine table separate; each is optimized against the other only through its interface section, never merged.

Scope: every want in the cell table's "wants" column. Each one is a UTM program: a record program swept over the body magnitudes, or a linear key swept over the voxels. Each has its own program module (like `residual_program` and `fingerprint_program`), its own `--run` part, and its own measurement. Ingest stays its own step and is never part of a run. The item numbers are the plan's own, retained . A number missing here is the engine workbook's (build_plan.md).

The bodies are one array: the tracker's leaves are the `max_tree` bodies in the order `flatten` packs them. A body's index is its frame's start plus its leaf.

Statuses: `proved`, `measured`, `built`, `theory`, `refuted`, `not so`. "Not built" means nothing of it exists yet.

4.1 DONE BEFORE THIS PLAN

item	what it is	status
two steps	`--ingest` alone; `--run <part>` repeated, in order	built
rows 2, 15: the print	`fingerprint_program`, 52 steps: the mass band and six signed golden bands of K′/m², K′ = s_as_b(m·M − S_aS_b), scales `voxel_pm`/gcd	built: 81,807 bodies of 44b6_0113de3b in 11.7 ms, CRC-64 `590f3a2c386647e3` twice

The whole tracker is written out as one equation, stage by stage, in the engine table's "The whole equation" (the aim is the entire equation, for algebra to improve the whole). Every item below that changes a stage updates that section too.

4.2 PHASE B: WANTS OVER THE BODY MAGNITUDES (RECORD PROGRAMS)

B1

ROW 7

WHAT IT IS a pair program gives each candidate's print cost (overlap triples and the climbed forward); `heaviest_matching` picks one to one and rewrites the forwards

STATUS built (`--print-match`, `print_pair`, FIELD_SIGNED added to read records back): a body against itself weighs the ceiling 1,275 on all 81,807, device equals host; in the tracker on 44b6_0113de3b 56,927 pairs swept at once in 217 us, 31,130 of 80,697 leaves matched, 19,113 forwards changed; edges unchanged at 38/12 of 50, because base.cfg's `unbound` linker picks by overlap weight and center cost and the forward is only one more candidate in its pool

B2

ROW 5

WHAT IT IS a pair program gives the centroid change as an exact rational; a three-body program predicts the landing by cross-multiplying; a tally counts agreement with the climb

STATUS built (`--velocity`, `velocity`, 42 steps, three members: two bodies and the climbed lag as its own array). The record is the velocity itself, the change $S_a'm - S_am'$ over $m \cdot m'$ per axis, and a truthy verdict per axis: the climbed lag is on a lattice point beside the dipole's change, $|\Delta - L| < 1$, cross-multiplied as $|R| < m \cdot m'$, written COMPARE·(COMPARE + 1). On 44b6_0113de3b: the check (each body and its climbed forward) 35,755 lanes, all three axes agree on 8,720 (z 22,660, y 13,227, x 13,306); the prediction (the last change against the next climb) 35,966 lanes, 752 agree (z 13,524, y 5,520, x 5,163); 0 differ from the host in both; edges unchanged at 38/12 of 50

B3

ROW 6

WHAT IT IS the membrane (4 nm) is a boundary that deforms. A body and its match overlap only in part, and the quantity to calculate is the size of that part. What stays inside the match under every jitter between the frames is the body's internal constituents; their shift is the body's velocity and vector; what falls out is the membrane deforming. Built with C1, the same measurement. And two membranes cannot pass through one another: for every touching pair A, B at t and their matches A′, B′ at t+1, the centroid difference retains its side, $(c_A - c_B) \cdot (c_A' - c_B') > 0$. Every mass denominator is positive, and the sign is exact from the cross-multiplied numerators: one pass writes each pair's difference as a record, and a second pass reads the two records and writes the truthy verdict. It is cheap here and needs no model

STATUS the deformation folded into C1. The no-crossing pair is built (`--contact-side`, `contact_side`). Pass 1, 23 steps, two bodies: $N = \sigma \circ (S_A\ m_B - S_B\ m_A)$, σ = voxel_pm over its common unit. Pass 2, 18 steps, reads two pass-1 records: dot = N·N′, c = COMPARE(dot + 1, 1), retained = c(c + 1), crossed = c(c − 1). The lanes are every touching leaf pair (`joined`) whose forwards both exist. Each pass is proved against the host. Ran on 44b6_0113de3b and

three 6bba: 0 touching pairs on every sample. The current bodies are separate components of the cut. None share a face, and `grow_leaves` writes `joined_count` = 0; E7 (`group_objects`) is inert too. What "touching" means for components of a cut is waiting on Doug (item 6)

B4

ROW 9

WHAT IT IS a three-body program (parent, child, sibling): band(m_c + m_s) = band(m_p), then $3\Delta^{T}K\Delta \geq \mathrm{tr}(K)\cdot|\Delta|^{2}$

STATUS built (`--division`, `division`, 100 steps, three members read through an index): the triples are every parent with two pieces whose backward climbs land on it. K is the parent's $m\cdot M - S\cdot S^{T}$ scaled by voxel_pm over its common unit, and Δ is the children's cross-multiplied centroid difference; both sides are homogeneous, and no mass is divided. On 44b6_0113de3b: 142,649 triples in 13.8 ms; mass conserved on 3,356, split along the parent's long axis on 99,155, both on 2,452; 0 differ from the host. That sample's ground truth holds no division. On 6bba_48816121, 6bba_09961292 and 6bba_cdcfe533: 326,836, 85,146 and 99,724 triples, 0 differ from the host. The three samples' ground truth holds 13 divisions; 3 put the parent and both children on three separate leaves, and 2 of those are among the triples. Mass holds on 1, the axis on 2, both on 1. Why the other 10 are lost (measured): both children are on one leaf. The cut at ℓ^{*} has not separated just-divided daughters; no node is off a leaf, and none falls outside consecutive tracked frames. So the division term's loss is row 1's cut, not the climb

B5

ROW 14

WHAT IT IS wherever two sizes are compared, the mass band is compared instead (size comparisons only; exact sums stay counts, since a band there would round)

STATUS built (`--mass-band`, `band_or_count` in golden_bands): `group_objects` (left/right bigger, leans), `arm_landing`, `focus_links`, `assign_bodies` (heaviest) and `track_bend` (both widest loops) compare bands when the flag is on and counts when it is off. Built clean; A/B run (D3): identical to counts on 24 of 24

B6

ROW 7, 8

WHAT IT IS OrganoidTracker 2.0's estimation made exact: each link's context-aware probability P(A|G) = Z_A / Z, the sum over every flow-conserving arrangement of the local subgraph (three steps out, divisions allowed) of the product of its weights, with and without link A. Theirs is exp(−E/T) of calibrated network energies. Ours (yes): a link weighs the fraction of the body that lands on the candidate, and "no link" the null draw's fraction; each body takes one outcome in every arrangement. The mass denominators cancel, and Z_A and Z are exact integers, compared by cross-multiplying, with no temperature and no calibration

STATUS built (`--marginal`, needs the null: `--null` with a floor; `marginal` module, device kernel one thread a body, host proof on orior's exact integer, 16 limbs). The local set G is the body, every source that shares one of its candidates, and their candidates, at one frame step. Each source takes one outcome and no two sources take the same target. A body whose set is too wide (more than 15 sources, 64 targets or 2^{16} arrangements) is asked alone. **Differs from the text above:** one step, not three, and no division arrangements (a source to two targets). Those are Doug's to decide. The tally compares the max probable in context, the heaviest candidate alone, and the climb's forward against the ground-truth edges. Measured on 44b6_0113de3b: 34,297 bodies asked in 50 ms (30,444 in context, 3,853 alone, 8 too wide), 123,538 options, device equal to the host. Of 49 ground-truth edges, the max probable in context is the truth for 16 and "no link" for 27. The heaviest candidate alone gets 38 and the climb's forward 35. The null's weight (the climbed null count) dominates T (counted at the drift); putting them on one footing is waiting on Doug (item 7)

4.3 PHASE C: WANTS OVER THE VOXELS AND THE HISTORY (LINEAR KEYS, THEN RECORD GATHERS)

C1

ROW 13, 6

WHAT IT IS the jitter scrub, with B3 (every width, retained per width, and the jitter between frames). For each body and its match at the climbed lag, a voxel stays at width w when every offset in the box of half-width w around the lag lands it inside the match. That is the match's Chebyshev distance from its own boundary, one exact integer per voxel. One pass buckets every voxel by the widest box it survives. Mass, sums and moments are re-summed per width: the core that stays is the internal constituents, its shift is the velocity, and what falls out at each width is the membrane deforming

STATUS built (`climb_machine_core`, `--core`), proved on 44b6_0113de3b: the width-0 core mass is C at the climbed lag on 35,755 of 35,755 forward climbers. Widths reach 5, and the per-width counts, masses and displacements are in row 13. The deeper the core, the less often its displacement lies within half a voxel of the climbed lag (62% at w=0, 42%, 30%, 13%, 13%). Still open: the core velocity feeding a link or the print, and contact κ_w (item 6) built on the same D

C2

ROW 9, 10

WHAT IT IS a running-sum key over the `.oapx` flips, then 8 corners gathered per body box: a bump that rises and settles is a division, one that rises and stays is a lysis

STATUS the measurement is built (`--box-history`, `box_history` module, `climb_machine_extents`). The box extents are stored per body from the stored runs; per window and bit,

one running-sum key is built on the device, and each body's box is read by its 8 corners. It is proved against walking every box: 0 of 11,780,208 differ on 44b6_0113de3b, 0 of 9,021,600 on 6bba_48816121 (its `.oapx` is written into the scratch set, not the data). Found on 6bba_48816121: every ground-truth division's parent is on a leaf of 0.65 to 1.0 M voxels whose box is the whole view, and 881 of 931 ground-truth nodes share a leaf with another ground-truth node (up to 13 on one leaf). So the bump cannot be read until row 1's cut holds one cell a leaf. The verdict is not built: it waits on the cut and on Doug (Waiting 14)

C3

ROW 4

WHAT IT IS a tuple sum over contacts gives each body's departure from the drift, weighted by mass; an origin program traces the flow back toward division and lysis events

STATUS not built

C4

ROW 12

WHAT IT IS every departure from the floor placed; the noise keys stamped in one cycle

STATUS not built; waits on Doug's design

C5, the residual as unit sweeps, is the engine's (the engine workbook's build plan).

4.4 PHASE D: PROVE AND MEASURE

D1 and D2 are the engine's (the engine workbook's build plan).

4.5 WAITING ON DOUG

1. Answered: `membrane_pm` is 4 nm; B3 is the membrane's deformation, built with C1.
2. The design for placing the floor departures and stamping the noise keys. (C4)
3. B1 rewrites the forwards as the plan says, but the linker base.cfg runs (`unbound`) does not choose by the forward. The print cannot move a link there. Should the print weigh in the linker's pool itself (row 7's "link each body to the one whose fingerprint agrees")?
4. Answered: B6's weights are the landing fraction and the null draw's fraction.
5. Ultrack [2] (Nature Methods 2025, the Royer lab's own tracker, the one this competition builds on) selects cells from a hierarchy of candidate segments jointly with the links: an ILP retains one node of every nested chain ($y_i + y_j \leq 1$ for $i \subset j$) and maximizes Σ IoU^4 over the links. Our component tree is that hierarchy already, and row 1's "one node per cell" is that selection. Their $\gamma = 4$ and event penalties are chosen numbers; an exact form would have to replace them.

6. Answered, touching: "we have assigned a boundary to these things and can use the overlap to estimate contact probability truthy falsy". Each body has an assigned boundary; two bodies are in contact where their boundaries overlap, and the overlap's size is the truthy magnitude. Built on C1's per-voxel distance: at width w, contact(A, B) = |{x : d_A(x) ≤ w ∧ d_B(x) ≤ w}|, where d is the Chebyshev distance to the body. Every width accumulates (item 8).

7. Answered, the null's weight: "we need the whole equation written out to figure out the exact transform of this term". From the equation: the null is the climb's own functional A_b(v*) taken in a null frame, and T is a different functional (one target, at the drift). The exact transform splits the same functional by target: T*(b, b′) = |{x ∈ b : L_{t+1}(x + v*_b) = b′}|, and Σ_{b′} T*(b, b′) = h(b). The links and "no link" then come from one operator: the real frame split by target, and the null frame whole.

8. Answered, C1's units: "everything should accumulate, everything should use the same atom class with their int cast into it". Every width's sums include every wider width, a running sum over the widths. Every quantity is carried as integers cast into the same atom lanes the engine sweeps, which puts the box on the atom's integer lattice, in voxels.

9. Answered, the cut: "It should be able to slide all the way to the end looking for coherence". Row 1's selection slides through every level of the component tree, to the top, and retains what stays coherent between frames. It does not cut the frame at one level. That is how a just-divided pair and everything else can each be retained at their own level.

10. B6's one frame step and the missing division arrangements (see B6).

11. The window of the one operator C: the climb's reach around the drift, or the whole view. Built: C at the 27 shifts around each climbed lag plus the drift (every read and proof needs only these). All three proofs hold on 44b6_0113de3b (retained count on 726,273 climbers, the local max on all of them, the overlap triples on 80,697 bodies). A wider window is only more cells.

12. "No link" in the marginal weighs the best null draw, max_k h^0_k. With T* (C at v*) the truth is picked for 19 of 49 ground-truth edges and "no link" wins 24. Should the draws enter as the best draw, as each draw its own "no link" outcome, or otherwise?

13. `train/44b6_551a5dba/44b6_551a5dba.iapx` proves in D3's counts run and fails its CRC in the bands run and on every later read: the rebuilt CRC is 6e6947161fd5c5cd against the file's 3245a85b0a90111e. Its modification time predates both runs, and three MD5 reads agree (e0c60eb2e06aa3eb618aec5b59599163). Nothing in this work writes the file. Re-ingest it, or check the drive?

14. C2's verdict: which of the 16 bits enter the bump (each separately, or the density Σ count·2^bit that the entropy cloud already uses), and how the 11-transition windows meet a one-frame division (the window holding the transition, against its neighbors?). Whatever the answer, it is only readable once the cut (item 9, the sliding cut) puts one cell on a leaf: on 6bba the dividing parents sit on leaves the size of the view. 9b. Answered, the sliding

cut: measure first. Slide ℓ through every level and report, per level, the component count and how many ground-truth cells share a leaf; no selection is built until the measurement is read. **Measured** (`--slide`, `max_tree_slide`): every level where the ground-truth cells' partition changes is found by bisection over the levels, one connected-component labeling a probe, using the fact that the partition only coarsens as ℓ falls. Every labeling's root count is proved against the tree's own count at that level (`counts[top − ℓ]`, from the max spanning forest): 0 of 1,102 differ on 44b6_0113de3b, and 0 of 26,454 on 6bba_48816121. On 6bba_48816121, 935 ground-truth cells over 100 frames: the current cut (the level with the max components, 626 a frame) holds 931 of them, and only **50** are alone on their leaf. At each frame's best single level, **903** are alone. At some level of its own chain, **925** are alone. Frame 0: the cut is at level 84,314 of 1,135,213, with the 4 ground-truth cells on shared leaves. At level 1,045,988 (204 components) all 4 are alone, and they stay separable down to 463,871. So the max-components level is low in the tree: many small pieces, with the cells merged into a few giant blobs. A per-frame or per-chain level separates almost every cell. 44b6_0113de3b has one ground-truth cell a frame, and every one is alone at every level (52 of 52). Edges are unchanged at 38/12 and 856/55/7, because nothing in the tracking reads the slide yet. Next is Doug's: the coherence criterion that picks the node per chain, now that the measurement is in.

9c. Answered, level pairing across frames: "Same residual value". Cut t and t+1 at the same exact residual r, and count the components of t that overlap exactly one component of t+1 at the frame's drift, and the reverse. **Measured** (`--overlap`, `max_tree_retain_frames` + `max_tree_overlap`):

- **Retained frames.** max_tree retains the last two frames' codes plus a code→residual table (9 limbs a code), built on the device from the first voxel of each run of equal residuals.
- **Threshold.** A residual r maps to each frame's threshold code by one device lane a code: the lane whose value is ≥ r while the value one code below is < r. Unused code entries are zero, and zero is below every admitted residual.
- **At each r:**
  - one connected-component labeling a frame;
  - each earlier voxel alive at r whose drifted place is inside the view and alive in the later frame emits the exact pair (earlier root, later root); voxels drifted outside the view emit nothing;
  - a device radix sort, then the distinct pairs, then partner counts a root;
  - "one" is the count of roots with exactly 1 partner, and "mutual" (AND = product) is the count of distinct pairs whose two ends both have exactly 1.
- **Sampled r values:** each frame's ground-truth partition events from the slide, plus its max-components cut, each mapped to its exact residual. This is a ground-truth-driven sample of the r values. Every r would be about 1.1M labellings a frame, and which r to sample is still Doug's.
- **6bba_48816121**, 99 pairs, 3,615 cuts:

  - at the earlier frame's cut: 61,679 components, 15,979 overlap exactly one (26%), and **6,688 mutual (11%)**;
  - the best mutual over each pair's sampled cuts sums to **13,783** (139 a pair, against 68 at the cut);
  - frame 0→1 at level 1,045,988: 204 components, 167 one, **144 mutual**. At the cut, level 84,314: 329 components, 83 one, 29 mutual;
  - the mutual count peaks high in the tree (codes about 1.0 to 1.1M of about 1.1M), at 150 to 200 components a frame, where 70–80% of the components are mutual. The max-components cut sits low, where 10–30% are.

- **44b6_0113de3b**, 53 pairs, 206 cuts, only two sampled levels a frame (the one ground-truth cell's appearance and the cut): at the cut 39,507 components, 14,193 one (36%), 7,962 mutual (20%); the best mutual sums to 8,166.

- **Cost and edges.** The overlap costs about 11 ms a cut: engines went from 125.2 s to 164.3 s on 6bba. Edges are unchanged at 856/55/7 and 38/12, because nothing reads it yet.

- **Next is Doug's:** the rule that replaces the max-components cut (for example, the r that maximizes mutual one-to-one, the product of the two verdicts), and whether r is swept over every value or over a data-driven sample.

9d. Direction: "make our cell tracker do all the cool stuff organoidtracker does". OrganoidTracker (vendored at `vendor_for_ref/OrganoidTracker`) is the feature list; each feature is rebuilt in our terms: exact integers, verdicts as fields, no weights or cutoffs. The mapping, with nothing built yet:

- **Detection.** Theirs: a U-Net predicts centers, then peak calling. Ours: max-tree bodies at the r the overlap picks (items 9, 9c).

- **Link score.** Theirs: a link network's probability per candidate pair. Ours: the exact overlap pairs at the drift (9c), with the shared-voxel count as the pair's integer field.

- **Division.** Theirs: a division network, then `pinpoint_divisions`. Ours: in the overlap graph, a component of t whose partners in t+1 are exactly two, each of them with exactly one partner back (1→2).

- **Appear and disappear.** Theirs: penalties with a probability floor (0.01) and an edge margin. Ours: 0→1 and 1→0 in the same graph, with view-edge contact read from the body's `touches` field instead of a margin.

- **Global linking.** Theirs: min-cost flow (dpct, Haubold 2016 [1]) with five chosen weights. Ours: min-cost flow on integer costs taken from the fields. Where each cost comes from is Doug's.

- **Gap bridging.** Theirs: `bridge_gaps`. Ours: the overlap across t→t+2 at the summed drift, for a 1→0 followed by a 0→1.

- **Error probabilities.** Theirs: local marginalization, which enumerates microstates in a 3-step neighborhood and weights them with Boltzmann factors at temperature 1.5. Ours: the same enumeration with exact counts; a link's field is (valid local solutions that contain it) / (all valid local solutions), retained as an exact pair.
- **Manual correction GUI and retraining loop.** Out of scope for the competition.
- **I/O, added: "Ctc geff api queries etc".**
  - Theirs: `imaging/geff_io.py` (`load_data_file` and `save_data_file`, with node and edge props), and `imaging/ctc_io.py` (reads and writes CTC: the man_track lineage file plus label images).
  - Ours today: `engine_geff_read` only, used for the ground truth. Ours to build: GEFF write of our tracks with node and edge props (the exact fields as props), and CTC read and write.
- **Query API, added by Doug.**
  - Theirs: `core/links.py`.
    - Positions: `find_futures`, `find_pasts`, `find_single_future`, `find_single_past`.
    - Appear and disappear: `find_appeared_positions`, `find_disappeared_positions`.
    - Tracks: `get_track`, `find_all_descending_tracks`, `find_all_tracks_in_same_lineage`, `find_starting_tracks`, `find_ending_tracks`, `find_all_tracks_in_time_point`.
    - Data attached to links and lineages: `get_link_data`, `find_all_links_with_data`, `get_lineage_data`.
  - Ours to build: the same queries over our track graph, in one module behind one request struct, with the exact fields as link data.
- **Order, ruled: the event census first.** `--overlap` classifies every component at each cut by its partner counts: 1→1 move, 1→2 division, 2→1 merge, 1→0 disappear, 0→1 appear, and the rest as n→m. It is measured against the ground truth's divisions on 44b6 and 6bba.
  - **Measured.**
    - Two device passes over the distinct (earlier root, later root) pairs:
      - "backs": for each root, how many of its partners have exactly one partner in return;
      - a census a root: 0 partners; 1 partner with 1 back (move); 2 partners with 2 backs (1→2, or 2→1 on the later side).
    - Mutual counted from the later side equals the earlier side's on every cut, 0 differ on both samples. That is the census's proof.
  - **At the earlier frame's cut:**

    - 6bba_48816121: 6,688 moves, 570 1→2, 590 2→1, 42,853 disappear, 40,996 appear, out of 61,679 components. The ground truth holds 5 divisions over these pairs.
    - 44b6_0113de3b: 7,962 moves, 819 1→2, 789 2→1, 22,545 disappear, 21,517 appear, out of 39,507. The ground truth holds 0 divisions.
    - So at max-components cut, about 70% of the components have no partner at the drift: specks that do not persist. Higher in the tree about 20% do.
    - The ground truth annotates about 9 cells a frame. Census totals cannot be graded against it. Only a per-division verdict can.
  - **Per ground-truth division, measured.** Each division is graded on a chain of verdicts, each implying the one before:
    - present: the parent at t and both daughters at t+1 are above the cut;
    - apart: the daughters are in two distinct components;
    - parted: the parent's component overlaps both of them at the drift, found by a device binary search in the sorted pairs;
    - caught: parted, and the parent's component is 1→2.
  - **6bba_48816121, 5 ground-truth divisions, taking each pair's best sampled cut.** At the max-components cut, 0 are parted.
    - 5 are present, 3 apart, 2 parted, 1 caught.
    - t=5→6: caught at level 1,056,677 (214 components, 162 mutual).
    - t=11→12: parted at 3 cuts but never 1→2, because the parent's component has more partners or its partners have more backs.
    - t=57→58: apart at 4 cuts but never parted. The parent's component at the frame's single drift does not reach both daughters.
    - t=16→17 and t=60→61: the daughters are never apart at any cut where the parent is also present.
  - **What that says:**
    - A just-divided pair separates only at levels where the parent is gone, or not at all.
    - One global drift misses daughters that move apart.
    - A division is found by a per-chain level (row 1's slide), with a per-body drift or climb in place of the frame's.
  - **Next is Doug's:** the rule for the cut; then GEFF and CTC write, and the query API (9d).

1. Ruled ("split it, go"): the slide and the overlap are in the engine as engine math (the engine workbook's build plan, item 24). Ground-truth cells, divisions and their grading (present, apart, parted, caught) are to be rebuilt in cell_tracking/src/slide/slide_score.cu from probes and links; that file holds the interface that precedes the probes. They cannot run until score_sample (engine/base/oracle) moves to cell_tracking, since score_sample's frame loop is where they hook in.

2. The driver through tessera. “The engines own runs need to go through tessera”. Done:

    - track_driver submits `--ingest` and each `--run` part as one job. The signum is BLAKE3(part name, 0, effective cfg). The declaration is the largest sample’s lanes × 2: `engine_source_lanes` (new, shape only) for ingest, the iapx head otherwise.
    - `--override` added. Holding 2 s, sweep 20 ms, idle 5 s. There is no path without tessera.
    - build_driver.sh builds and publishes tessera_daemon beside the driver.
    - Run on a scratch set: ingest declared 838,860,800 with a peak of 5,091,037,184; prove had peaks of 3,962,761,216 and 3,958,566,912. The history is 128 bytes.
    - Finding, not changed (A11 is Doug’s rule): a job is reserved at its declaration even when its retained peak is larger. Reserving max(declared, retained peak) is the fix.

3. **The score.** Tessera and portability stop here (“stop getting bogged down in portability”).

    - **Baseline on the current engine: SCORE 0.209** (object, largest 400; Jaccard 0.210, 961,081 nodes against 880,906). The engine with peak-basin nodes scores **0.661** (basin, largest 400; Jaccard 0.670). The regression comes from the v2 split.
    - Cause: the peak-basin nodes are peak basins (`binomial_basins`/`peak_basins`, steepest ascent over 26 neighbors, membership where the residual is positive), about 300,000 leaves a sample grouped into objects. The v2 engine’s nodes are max-tree components at max-components level, one object a leaf, cells merged into blobs. No basin module is in the engine, and nothing in it does steepest ascent.
    - Built: `--basins` (rule → EngineBuffers → EngineBodiesRequest → MaxTreeObjectsRequest.basins). `max_tree_basin_ascent_kernel` points every voxel at the highest of itself and its 26 neighbors (key code, then ~index), and `max_tree_cc_flatten_kernel` follows each positive voxel to its peak; the peak, mark and body kernels are unchanged. The level cut stays the default until basins score.

5

LEDGER

Purpose: Every measurement, in the order it was taken, with the samples it ran on, the number, and what it settled. No result is taken twice and none is quoted without its run.

5.1 2026-09-21

5.1.1 *The driver split*

what	samples	result	settles
the split driver (22 functionals, now under `engine/` and `cell_tracking/src/`) against the driver before the split	44b6_0113de3b, 44b6_0b24845f	edges, object files and every score line byte identical	the split changed nothing (proved)

5.2 2026-09-22

5.2.1 *The module move and the root layout*

what	samples	result	settles
the driver from the new layout, run from the root, against the driver before the split	44b6_0113de3b, 44b6_0b24845f	edges and score rows identical; object files identical but for the embedded .cfg naming the new paths	the move changed nothing (proved)

5.2.2 *The competition metric*

what	samples	result	settles
components mode dump (an older mode), best policy (the 400 largest per frame)	13 of the 44b6	SCORE 0.2376; edge Jaccard 0.238; TP 713, FP 467, FN 1,815	the internal 98% does not carry over to the metric

what	samples	result	settles
same dump, every node	13 of the 44b6	60.4% of ground-truth edges lost because the source ground-truth node has no predicted node within 7 µm; 1.5% land elsewhere	that dump's loss is detection instead of linking; the dump is an older mode and must be redone on the current engine

5.2.3 *The 6bba samples*

The training set is 71 samples of 44b6 and 128 of 6bba; the test set holds both. Every earlier row is on 44b6. 6bba's ground truth is far denser (345 to 1,183 edges a sample, against about 50 to 270 for 44b6), and under micro averaging 6bba carries the majority of the score.

what	samples	result	settles
the current engine, internal count	6bba_05b6850b, 05dbofb1, 062c8d37, 07477033, 07e24132	3,756 of 3,873 edges correct, 97.0%; 115 wrong; 2 end-points undetected	the engine runs on 6bba as it is
the same run under the metric's matching, every node	the same five	78.5% of ground-truth edges hit; 1.2% source with no match; 19.6% land elsewhere, a median 6.7 µm from the target's match	on 6bba the loss is a link landing on a neighboring node: the node set is far finer than the cells, and bodies, not finer pieces, are what should be linked

5.2.4 *44b6 on the current engine*

what	samples	result	settles
internal count	the 25	6,254 of 6,358 edges correct, 98.4%; 104 wrong	
the metric's matching, every node	the 25	87.1% of ground-truth edges hit; 1.1% source with no match; 10.9% land elsewhere, a median 5.75 µm off	the older components dump's 60% with no match was that dump, not the engine
SCORE, the 400 largest nodes a frame	the 25	0.661: edge Jaccard 0.670, node count factor 0.986 (1,000,000 nodes against 880,906 estimated)	the node count is the lever: every node gives Jaccard 0.635 but 5.8 million nodes, and the score falls to 0.279

6

SCAN, THEN SORT

Purpose: The rule that orders the driver: every frame of every sample is scanned first, and sorting happens only at the end.

Scope: the driver's stages, So to S11 of cell_tracking_table.md.

Ruled and said more than once: "you scan everything and then you sort stuff. You don't need to sort it before you scan in all of the samples."

- The driver scans first: every frame of every sample gives points and fields that bound nothing (peaks as points, the drift, the entropy history).
- Sorting (linking, retention, divisions, any partition into regions) happens only at the end, over the whole sample, and anything set-wide over every sample.
- Never partition a frame into regions (watershed/toboggan catchments, a max-tree cut) as it is read: that decides membership from one frame. Doug dislikes "basin" because the name, and the thing, bound the set.
- The score reads only node positions (7 µm) and edges; regions are not needed for it.
- The driver's stages are the table "The driver: scan everything, sort at the end" in cell_tracking_table.md (S1 to S11). OrganoidTracker's techniques (O1 to O22) and Hawkins et al. 2025's fates (H1 to H6) are mapped onto those stages there.
- The rule is Doug's.

7

WALK BACK

Purpose: The rule every tracker stage must satisfy: it can be walked back to what it came from.

Scope: every stage of the driver (cell_tracking_table.md, its driver section).

Ruled: “The whole point of the tracker is to track cells. If you can’t walk back something you did you built it wrong.”

- Every stage retains what it came from: a point retains its frame, voxel and exact level; a transform retains its exact inverse; a link retains its two points and its evidence; a choice in the sort retains the options it passed over, with their weights.
- Nothing is deleted. What the sort does not retain is marked, and the mark can be walked back.
- A map that sends two readings to one (a clip, a rounding, a bin) cannot be walked back and is not built.
- An example: OrganoidTracker’s 1%/99% intensity clip sends many readings to one and fails the rule. So holds a per-sample affine footing as a pair instead, plus the set’s cumulative count as the contrast, and both are invertible.
- It is written into the driver section of cell_tracking_table.md.

8

BUILD IT RIGHT FIRST

Purpose: The rule against building an interim version already known to need redoing.

Scope: every build of the cell program.

Ruled: build it right from the first build, and stop building it wrong. His analogy: a contractor who builds it wrong, gets paid, then gets paid again to renovate.

- If the right design is known (or Doug has just ruled it), build that directly. Do not build, prove or run a stopgap on the old path to "validate the pipeline first" and then redo it.
- Work out the whole design against the rules (walk back, scan-then-sort, no chosen numbers, any width) before writing code. The first build is then the build that stays.

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